Dynamic Oligopoly and Innovation

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Abstract

We develop and estimate a many-firm endogenous growth model in which firms make forward-looking R&D decisions while interacting through empirically measured product-market rivalry and technology-spillover networks. The model's linear-quadratic structure reduces the dynamic oligopoly to a coupled system of Riccati equations, which we solve for the 2017 cross section of 757 publicly listed U.S. patenting firms. Gaps between social and private returns to R&D are large and heterogeneous: the median social-to-private return ratio is 1.30. A planner who controls R&D but holds product-market allocations fixed raises R&D to 231% of the competitive level, yielding a consumption-equivalent welfare gain of 0.50%. Despite wedge heterogeneity, a 33% uniform R&D subsidy delivers a 0.45% welfare gain, because the constrained planner's gain comes mainly from expanding aggregate R&D rather than reallocating it. Substantially larger gains, 8.97%, arise only when product-market allocations are reallocated as well, a margin beyond the reach of R&D subsidies.

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1 Introduction

Product-market competition and research and development (R&D) allocations are jointly determined. Firms' R&D decisions shape future productivity, product quality, and competitive positions, while current product-market rivalry feeds back into profits and the incentives to invest in R&D. The network-mediated part of these interactions operates through two pairwise channels: business stealing, which depresses competitors' profits, and technology spillovers, which raise other firms' productivity. In addition, firms do not generally appropriate the full household value generated by their innovations. Because these forces are heterogeneous across firms and firm pairs, the social-private R&D wedge is firm specific.

The two networks need not coincide: firms that compete closely in product markets may be distant in technology space, and firms that generate large spillovers may not be close product-market rivals. Yet common innovation-policy instruments, such as broad R&D subsidies and tax credits, change the price of R&D by the same proportion for every firm. The central question is which innovation margins such instruments can address, and which margins require information about firms' positions in the product-market rivalry and technology-spillover networks.

Existing endogenous growth models typically abstract from this firm-pair heterogeneity. Some models assume monopolistic competition without strategic interaction, while others allow strategic interaction among only a few firms. The difficulty is computational. In oligopoly, forward-looking R&D choices cause the dimensionality of the dynamic problem to increase rapidly with the number of strategically interacting firms. Existing quantitative frameworks therefore have had difficulty combining forward-looking strategic interaction among many firms with empirically measured product-market rivalry and technology-spillover networks. As a result, we lack firm-level measures of the social-private R&D wedges that policy should target.

This paper develops and estimates an endogenous growth model in which hundreds of firms make forward-looking R&D decisions while interacting through two empirically measured networks: a product-market rivalry network and a technology-spillover network. The contribution is twofold. Methodologically, the model's linear-quadratic structure makes a many-firm dynamic oligopoly tractable: equilibrium is characterized by a coupled system of algebraic Riccati equations rather than a high-dimensional dynamic program, so we can solve the game for the 2017 cross section of 757 U.S. publicly listed patenting firms while preserving a continuous state space and the complete firm-to-firm networks. Substantively, the estimated model delivers firm-level social-private R&D wedges and a welfare decomposition with a sharp policy message: despite large and heterogeneous wedges, a 33% uniform subsidy captures most of the gain available to a planner who controls R&D alone, raising consumption-equivalent welfare by 0.45% compared with the planner's 0.50%, while the far larger gain under the full social-planner allocation, 8.97%, comes from reallocating product-market outcomes, a margin beyond the reach of R&D subsidies.

The model extends Pellegrino (2025)'s static product-market framework into a dynamic setting with knowledge accumulation, R&D effort, and technology spillovers. Conditional on the current knowledge-capital vector, a static product-market game under generalized hedonic linear demand maps firms' knowledge capital into quantities and profits; its key property is that gross profits are quadratic in knowledge capital. Knowledge capital accumulates linearly through own R&D effort and through the technology-spillover network. The dynamic R&D game is therefore linear-quadratic: value functions are quadratic in the state, Markov strategies are linear in it, and computation scales polynomially rather than exponentially in the number of firms. In the deterministic version, a technology transition matrix summarizes spillovers, obsolescence, and endogenous R&D; when a positive long-run eigenvector exists, the same matrix characterizes balanced-growth implications.

We estimate the framework on U.S. publicly listed firms with patents over 1989–2019. Following Pellegrino (2025), we measure product-market rivalry using the product-similarity data of Hoberg and Phillips (2016); following Bloom et al. (2013) and Lucking et al. (2019), we measure technological proximity from patent-classification overlap using the DISCERN patent-firm link (Arora et al., 2024). Given these networks, the model's static equilibrium conditions recover firm-level knowledge capital from Compustat accounting data. The spillover intensity is estimated from the panel law of motion of recovered knowledge capital, yielding a rounded central estimate of $\beta = 0.02$. The R&D efficiency and depreciation parameters are jointly calibrated to aggregate R&D intensity and observed-state output growth. The model fits untargeted moments well: the correlation between model-implied and observed log sales is 0.96, the corresponding correlation for log R&D expenditure is 0.72, and model-implied knowledge growth evaluated at the 2010 state predicts realized 2010–2017 knowledge growth with a correlation of 0.58.

The estimated wedges are large and heterogeneous. At the observed 2017 knowledge-capital distribution, the median ratio of social to private marginal returns to R&D is 1.30, and social returns exceed private returns for 82.1% of firms with positive private and social marginal returns, consistent with aggregate underprovision. The dispersion is substantial: 22.3% of firms have ratios above 1.5, while for firms in the bottom tail the local first-order-condition measure points toward taxing rather than subsidizing marginal R&D.

This dispersion suggests that a uniform instrument should leave much of the attainable R&D-policy gain on the table. An exact decomposition traces each firm's wedge to the net of two opposing forces: a positive appropriability component, reflecting household value the firm creates but does not capture as producer value, and a negative rival-profit component operating through business stealing.

To separate the policy margins, we compare the competitive equilibrium with planner and monopoly counterfactuals. A constrained planner who holds the competitive product-market allocation fixed and chooses only the R&D rule raises total R&D to about 231% of the competitive level, increasing output growth by 0.28 percentage points and consumption-equivalent welfare by 0.50%. The corresponding constrained monopolist, who internalizes the rival-profit force but not the appropriability force, instead cuts R&D roughly in half and lowers welfare by 1.09%. This constrained case can also be read as R&D-only common control, or an economy-wide research joint venture, while product-market allocations remain competitive. Allowing production and R&D to be reallocated jointly multiplies the gain: the full social-planner allocation raises welfare by 8.97%. Most of the planner's attainable welfare gain thus lies not in the R&D margin itself but in the product-market allocation that R&D policy takes as given.

We then evaluate an implementable policy: a uniform R&D subsidy financed by lump-sum taxes. Household welfare peaks at a subsidy rate of 33%, which raises output growth by 0.27 percentage points, compared with 0.28 under the constrained planner, and welfare by 0.45%, compared with 0.50%. Despite wedge heterogeneity, a uniform instrument captures roughly ninety percent of the gain attainable through R&D policy alone. The remaining fixed-product-market welfare gap is small: the planner's advantage comes mainly from expanding aggregate R&D rather than from reallocating it across firms. The margins are therefore ordered. The aggregate R&D margin is large and well served by a uniform instrument; the remaining firm-level targeting margin is small in this comparison; and the product-market margin is by far the largest but calls for instruments other than R&D policy.

This paper contributes to three strands of the literature. First, it contributes to work on innovation incentives, appropriability, business stealing, technology spillovers, and innovation networks (Arrow, 1962; Spence, 1984; d'Aspremont and Jacquemin, 1988; Aghion et al., 2001, 2005; Acemoglu and Akcigit, 2012; Peters, 2020; Akcigit and Ates, 2021, 2023; De Ridder, 2024). Closest to our mechanism, Bloom et al. (2013) show empirically that product-market proximity and technological proximity have opposite effects on innovation incentives. Closest on the policy-network margin, König et al. (2019) analyze R&D alliance networks in which collaborations lower production costs while firms compete in product markets, and characterize welfare-maximizing R&D subsidies. Liu and Ma (2026) study how innovation networks shape the allocation of R&D across connected firms, and Cavenaile et al. (2026) develop an oligopolistic general-equilibrium Schumpeterian model with strategic interaction among a small number of firms. Relative to this literature, we embed product-market rivalry and technology spillovers in a many-firm dynamic oligopoly disciplined by firm-level and firm-pair-level data. This equilibrium structure lets us quantify the distribution of social-private R&D wedges, connect them to welfare, and distinguish the policy margins addressed by uniform subsidies from those tied to product-market reallocation.

Second, the paper relates to the dynamic oligopoly literature in empirical industrial organization (Ericson and Pakes, 1995; Besanko and Doraszelski, 2004; Doraszelski and Pakes, 2007; Weintraub et al., 2008). This literature develops Markov perfect industry dynamics and methods for handling large state spaces. We differ by exploiting a linear-quadratic structure that makes a full-scale many-firm continuous-state R&D game tractable with empirically measured product-market rivalry and technology-spillover networks.

Third, the paper contributes to research that incorporates oligopolistic competition into macroeconomic models (Neary, 2003; Atkeson and Burstein, 2008; Azar and Vives, 2021; Pellegrino, 2025; Ederer and Pellegrino, 2026; Bustamante and Pellegrino, 2026). Our product-market environment builds on Pellegrino (2025) and Ederer and Pellegrino (2026). Closely related contemporaneous work by Bustamante and Pellegrino (2026) develops a Network-Q model of corporate investment in oligopolistic product markets using the same GHL demand system. Their focus is neoclassical capital investment, firm valuation, markups, and merger counterfactuals; we instead add a technology-spillover network and forward-looking R&D investment to study how product-market rivalry and technology spillovers jointly shape growth, welfare, and innovation policy.

The remainder of the paper proceeds as follows. Section 2 constructs an endogenous growth model incorporating product-market rivalry and technology-spillover networks. Section 3 describes the data and estimation strategy. Section 4 reports the wedge measures, counterfactual allocations, and the uniform-subsidy evaluation. Section 5 concludes.

2 Model

We develop an endogenous growth model in which granular firms strategically interact in both product and technology spaces to examine the implications for firms' dynamic R&D effort, economic growth, and household welfare.

2.1 Environment

Our environment incorporates two networks: a product-market rivalry network and a technology-spillover network. There are n oligopolistic firms, indexed by $i \in \{1, 2, \ldots, n\}$; they choose output and dynamic R&D effort while interacting strategically through these networks. We interpret each firm as supplying one composite differentiated product, so firm and product indices coincide throughout. The set of firms, product-characteristic vectors, and technological-proximity matrix are fixed over time. Innovation changes firms' knowledge capital, labor productivity, and product quality, but not their locations in product or technology space. Time is continuous and infinite, $t \in [0, \infty)$.

2.1.1 Demand

We use the Generalized Hedonic Linear (GHL) demand system of Pellegrino (2025) to model product-market rivalry among many oligopolistic firms. In reduced form, the final-good aggregator is

$$Y_t = \boldsymbol{q}_t^T \boldsymbol{b}_t - \frac{1}{2} \boldsymbol{q}_t^T \boldsymbol{\Sigma} \boldsymbol{q}_t \tag{1}$$

where $\boldsymbol{q}_t \in \mathbb{R}^n$ is the vector of product quantities and $\boldsymbol{b}_t \in \mathbb{R}^n$ is the vector of product-level qualities. The product-rivalry matrix is

$$\boldsymbol{\Sigma} \equiv \alpha \boldsymbol{S} + (1 - \alpha) \boldsymbol{I},$$

where $\boldsymbol{S}$ is the product-similarity matrix implied by the GHL characteristic structure, $\boldsymbol{I}$ is the identity matrix, and $\alpha \in [0, 1]$ maps product similarity into substitutability. In the underlying GHL aggregator, α is the weight placed on common characteristics rather than idiosyncratic product characteristics, so higher α makes common-characteristic overlap translate more strongly into product-market substitutability. We normalize $S_{ii} = 1$, so $\sigma_{ii} = 1$ for every firm, and for $i \neq j$, $\sigma_{ij} = \alpha S_{ij}$. From Equation (1), a larger σ_{ij} implies a larger quadratic penalty when q_i and q_j are both high, so σ_{ij} measures the strength of product-market rivalry between firms i and j. The GHL characteristic-level construction underlying $\boldsymbol{S}$ and the reduced-form aggregator in Equation (1) is collected in Section A.1.

2.1.2 Technology

Each firm has a linear production technology in labor:

$$q_{i,t} = a_{i,t} l_{i,t} \tag{2}$$

where $a_{i,t}$ is the labor productivity of firm i at time t and $l_{i,t}$ is the production labor employed by firm i at time t.

Firms possess their own knowledge capital and use it to improve labor productivity and product quality. As shown in Equation (1), $b_{i,t}$ is the value (in final-good units) generated by the first unit of product i, so we interpret $b_{i,t}$ as product quality. Firms allocate knowledge capital between labor productivity $a_{i,t}$ and product quality $b_{i,t}$:

$$\zeta a_{i,t} + b_{i,t} = z_{i,t} \tag{3}$$

where $\zeta > 0$ governs the trade-off between improving labor productivity and product quality[1].

We model the law of motion for knowledge capital, which rises through technology spillovers from technologically similar firms and through the firm's own R&D effort. Spillovers operate through source firms' knowledge capital rather than directly through their contemporaneous R&D expenditures; a firm's R&D affects other firms only by raising its future knowledge capital. Let $\tilde{\mathbf{\Omega}} \in \mathbb{R}^{n \times n}$ denote recipient-normalized technology-spillover exposure, with entries $\tilde{\omega}_{ij}$. We use a row-receiver convention: row i describes the sources of spillovers received by firm i. Thus, $\tilde{\omega}_{ij} \geq 0$ is the weight placed on source firm j in firm i's spillover exposure. We set the diagonal to zero, $\tilde{\omega}_{ii} = 0$, and rows with positive off-diagonal exposure are normalized so that $\sum_{j \neq i} \tilde{\omega}_{ij} = 1$. The effective spillover matrix is $\mathbf{\Omega} = \beta \tilde{\mathbf{\Omega}}$, where $\beta > 0$ scales these relative exposure weights into spillover intensities.

Let $\boldsymbol{x}_t = (x_{1,t}, \ldots, x_{n,t})^T \in \mathbb{R}^n$ denote R&D efforts, and let $\boldsymbol{W}_t = (W_{1,t}, \ldots, W_{n,t})^T$ denote independent standard Wiener processes across firms. The control $x_{i,t}$ is an R&D effort that raises knowledge capital linearly, while the resource cost of generating this effort is quadratic, as specified in the resource constraint below. Knowledge capital follows a multi-dimensional geometric Brownian motion:

$$d\boldsymbol{z}_t = \left((\mathbf{\Omega} - \delta \boldsymbol{I})\, \boldsymbol{z}_t + \mu \boldsymbol{x}_t \right) dt + \gamma \operatorname{diag}(\boldsymbol{z}_t)\, d\boldsymbol{W}_t \tag{4}$$

where $\mu > 0$, $\delta > 0$, and $\gamma \geq 0$. The parameter μ governs the efficiency of R&D, γ the magnitude of shocks, and δ the depreciation rate of knowledge capital.

[1] Assuming optimal allocation of knowledge capital between labor-productivity and quality improvements is necessary for a balanced-growth-path equilibrium. Similarly, Grossman et al. (2017) and Jones and Liu (2024) obtain a balanced growth path by assuming two productivity-enhancing technologies (capital and human capital; capital productivity and automation, respectively) and allocating resources between them.

2.1.3 Resources and Household Preferences

We close the model with resource constraints and representative household preferences. The representative household supplies labor, owns firms, and consumes the final good. Production labor is supplied inelastically:

$$L = \sum_i l_{i,t}. \tag{5}$$

The final good is used for consumption and R&D investment. Following the endogenous growth literature (e.g. Acemoglu et al. 2016), we assume that the innovation elasticity for each firm is 0.5. Equivalently, the resource cost of firm-level R&D effort is quadratic:

$$C_t + \sum_i x_{i,t}^2 = Y_t$$

where C_t is consumption of the final good at time t. Finally, we assume that the representative household is risk-neutral and discounts the future at rate $\rho > 0$:

$$\mathbb{E}_0 \left[\int_0^\infty \exp\left(-\rho t\right) C_t \, dt \right]$$

2.2 Competitive Equilibrium

We characterize a linear Markov perfect equilibrium. The key state variable is the vector of knowledge capital $\boldsymbol{z}$. Given $\boldsymbol{z}$, the static product-market game determines quantities, technology allocations, prices, and the real wage. Firms then choose Markov R&D policies that affect future knowledge capital through Equation (4).

Definition 1 (Competitive equilibrium)**.** A competitive equilibrium consists of Markov R&D policies $\{x_i(\boldsymbol{z})\}_{i=1}^n$, firm value functions $\{V^i(\boldsymbol{z})\}_{i=1}^n$, static allocations $\boldsymbol{q}(\boldsymbol{z})$, $\boldsymbol{a}(\boldsymbol{z})$, $\boldsymbol{b}(\boldsymbol{z})$, the real wage $w(\boldsymbol{z})/P$, and relative prices $\boldsymbol{p}(\boldsymbol{z})/P$ such that: (i) the final-good producer maximizes profits; (ii) firms play an interior static Cournot equilibrium given $\boldsymbol{z}$; (iii) each firm's R&D policy is a Markov best response in the dynamic problem; (iv) knowledge capital follows Equation (4); and (v) the labor market clears.

This structure lets us separate the problem into static and dynamic components. Conditional on the current knowledge-capital vector $\boldsymbol{z}_t$, the production and technology-allocation choices determine current quantities, prices, and gross profits but do not directly affect future knowledge capital. R&D affects current payoffs through its flow cost and affects future payoffs only through the law of motion for $\boldsymbol{z}_t$. Thus, we first solve the static product-market game as a mapping from knowledge capital into profits, and then use this mapping as the payoff flow in the dynamic R&D game.

2.2.1 Final-Good Producer's Problem

The final good is produced competitively. In each period, the final-good producer chooses the quantity of each product that maximizes profits, given prices.

$$\max_{\boldsymbol{q}_t} P_t Y_t - \boldsymbol{q}_t^T \boldsymbol{p}_t$$

where $\boldsymbol{p}_t$ is the vector of product prices at time t and P_t is the final-good price at time t. The final-good producer's profit maximization problem yields the following linear inverse demand function:

$$\frac{\boldsymbol{p}_t}{P_t} = \boldsymbol{b}_t - \boldsymbol{\Sigma}\boldsymbol{q}_t \tag{6}$$

2.2.2 Static Firm Problem

We assume that firms maximize real profits, so allocations are independent of the choice of numeraire (Azar and Vives, 2021). We characterize the interior static Cournot-Nash equilibrium in which each firm takes the real wage and rivals' quantities as given when choosing output and the allocation of knowledge capital between labor productivity and product quality. The static block maps the knowledge-capital vector into quantities:

$$\boldsymbol{q}_t = \boldsymbol{N}\boldsymbol{z}_t \tag{7}$$

where

$$\boldsymbol{N} \equiv \left(2\frac{\zeta}{L}\boldsymbol{J} + \boldsymbol{\Sigma} + \boldsymbol{I}\right)^{-1},$$

and $\boldsymbol{J}$ is an $n \times n$ matrix whose entries are all one. The first term inside the inverse, $2\zeta\boldsymbol{J}/L$, captures the labor-cost channel, while $\boldsymbol{\Sigma}$ captures product-market rivalry. The additional $\boldsymbol{I}$ is the Cournot own-output strategic term: $\boldsymbol{\Sigma}\boldsymbol{q}$ enters inverse demand, and the derivative of firm i's revenue with respect to q_i adds another $\sigma_{ii}q_i = q_i$ under the normalization $\sigma_{ii} = 1$. The derivation is in Section A.2. Lemma 1 shows that the inverse is well defined and that the interior static Cournot block is uniquely pinned down for each knowledge-capital vector.

Let $\pi_{i,t}$ denote firm i's real gross profit before subtracting R&D costs. In the interior

Cournot equilibrium, the normalization $\sigma_{ii} = 1$ implies that static gross profit equals $q_{i,t}^2$. Hence the static payoff entering the dynamic R&D game is quadratic in knowledge capital:

$$\pi_{i,t} = q_{i,t}^2 = \boldsymbol{z}_t^T \boldsymbol{N}_i^T \boldsymbol{N}_i \boldsymbol{z}_t \equiv \boldsymbol{z}_t^T \boldsymbol{Q}_i \boldsymbol{z}_t$$

where $\boldsymbol{N}_i$ denotes the i-th row of $\boldsymbol{N}$, and $\boldsymbol{Q}_i \equiv \boldsymbol{N}_i^T \boldsymbol{N}_i$. This quadratic payoff representation is the static source of the model's linear-quadratic dynamic structure.

2.2.3 Dynamic R&D Game

Given a static production strategy profile, we consider a dynamic R&D game. In the dynamic game, given other players' strategies $\left\{x_{j,t}\right\}_{j\neq i, t\geq 0}$, firm i chooses R&D effort $\{x_{i,t}\}_{t\geq 0}$ to maximize the expected discounted present value of profits:

$$\max_{\left\{x_{i,t}\right\}_{t\geq 0}} \quad V^i\left(\boldsymbol{z}_0\right) \equiv \mathbb{E}_0\left[\int_0^\infty \exp\left(-\rho t\right)\left\{\pi_{i,t} - x_{i,t}^2\right\} dt\right]$$

while the state evolves according to Equation (4). We use the fact that the interest rate is pinned down by the household's Euler equation, $r_t = \rho$. In firm i's HJB equation, x_i is the choice variable and the other components of $\boldsymbol{x}$ are evaluated at rivals' Markov strategies. The dynamic problem gives the following HJB equation:

$$\begin{aligned}\rho V^i\left(\boldsymbol{z}\right) = \max_{x_i}&\left\{\boldsymbol{z}^T \boldsymbol{Q}_i \boldsymbol{z} - x_i^2 + V_{\boldsymbol{z}}^i\left(\boldsymbol{z}\right)\left[\left(\boldsymbol{\Omega} - \delta \boldsymbol{I}\right)\boldsymbol{z} + \mu \boldsymbol{x}\right]\right.\\ &\left. + \frac{\gamma^2}{2}\mathrm{Tr}\left[\mathrm{diag}\left(\boldsymbol{z}\right) V_{\boldsymbol{z}\boldsymbol{z}}^i\left(\boldsymbol{z}\right)\mathrm{diag}\left(\boldsymbol{z}\right)\right]\right\}.\end{aligned} \tag{8}$$

The first two terms are firm i's net instantaneous profit, while the last two terms capture expected changes in continuation value from knowledge accumulation and idiosyncratic shocks.

This dynamic game is a linear-quadratic differential game, which significantly simplifies the analysis. The simplification follows from a standard guess-and-verify argument. Guess that the value can be expressed as a quadratic form of knowledge capital:

$$V^i\left(\boldsymbol{z}\right) = \boldsymbol{z}^T \boldsymbol{X}^i \boldsymbol{z}. \tag{9}$$

Because $\boldsymbol{z}^T \boldsymbol{X}^i \boldsymbol{z}$ depends only on the symmetric part of $\boldsymbol{X}^i$, we take $\boldsymbol{X}^i$ to be symmetric without loss of generality. The derivative algebra is collected in Section A.4.

In a linear-quadratic differential game, firms' strategies are expressed as linear functions of the state variables. Let $\boldsymbol{X}_i^i$ denote the i-th column of $\boldsymbol{X}^i$. The first-order condition with respect to x_i gives the linear Markov strategy

$$x_i = \left(\mu \boldsymbol{X}_i^i\right)^T \boldsymbol{z}. \tag{10}$$

This rule is the unconstrained linear-quadratic feedback policy, which we use as a tractable interior benchmark.

Given firms' linear strategies, the law of motion for knowledge capital $\boldsymbol{z}_t$ can be rewritten as a geometric Brownian motion with constant coefficients. Define $\widetilde{\boldsymbol{X}}$ as the matrix obtained by stacking $\boldsymbol{X}_i^i$:

$$\widetilde{\boldsymbol{X}} \equiv \begin{bmatrix} \boldsymbol{X}_1^1 & \cdots & \boldsymbol{X}_n^n \end{bmatrix}^T$$

Then, the law of motion is rewritten as

$$d\boldsymbol{z}_t = \boldsymbol{\Phi} \boldsymbol{z}_t dt + \gamma \operatorname{diag}(\boldsymbol{z}_t)\, d\boldsymbol{W}_t$$

where the drift matrix $\boldsymbol{\Phi}$ is defined by

$$\boldsymbol{\Phi} \equiv \boldsymbol{\Omega} - \delta \boldsymbol{I} + \mu^2 \widetilde{\boldsymbol{X}}.$$

We refer to the matrix $\boldsymbol{\Phi}$ as the *technology transition matrix*. It consists of the exogenous technology-spillover matrix $\boldsymbol{\Omega}$, depreciation $-\delta \boldsymbol{I}$, and the endogenous R&D matrix $\mu^2 \widetilde{\boldsymbol{X}}$.

For any square matrix $\boldsymbol{A}$, let $\mathcal{D}(\boldsymbol{A})$ denote the diagonal matrix formed from the diagonal entries of $\boldsymbol{A}$. With independent multiplicative shocks, the diffusion term in Equation (8) contributes $\gamma^2 \mathcal{D}(\boldsymbol{X}^i)$ to the quadratic coefficient. The HJB equation can be transformed into an algebraic Riccati equation, making the many-firm dynamic oligopoly problem computationally feasible. Substituting the quadratic value function and the linear policy Equation (10) into Equation (8), and collecting quadratic terms, we obtain stacked algebraic Riccati equations

$$0 = \boldsymbol{Q}_i - \mu^2 \boldsymbol{X}_i^i \left(\boldsymbol{X}_i^i\right)^T + \left(\boldsymbol{\Phi} - \frac{\rho}{2}\boldsymbol{I}\right)^T \boldsymbol{X}^i + \boldsymbol{X}^i \left(\boldsymbol{\Phi} - \frac{\rho}{2}\boldsymbol{I}\right) + \gamma^2 \mathcal{D}\left(\boldsymbol{X}^i\right) \tag{11}$$

for $i = 1, 2, \ldots, n$. Note that the terms $\mu^2 \boldsymbol{X}_i^i \left(\boldsymbol{X}_i^i\right)^T$ and $\boldsymbol{\Phi}$ depend on the $\boldsymbol{X}^j$ of other firms $j \neq i$. Therefore, to solve for $\boldsymbol{X}^i$, we need to simultaneously solve the algebraic Riccati equations for all firms.

We now define stability of the solution.

Definition 2. The solution of the stacked algebraic Riccati Equation (11) is *stabilizing* if it delivers finite expected discounted payoffs and satisfies the transversality condition

$$\lim_{T\to\infty} \mathbb{E}_0 \left[\exp(-\rho T) V^i(z_T)\right] = 0$$

for all firms i and admissible initial states.

We keep the shock term in the model equations to state the stochastic LQ extension, but all quantitative exercises in the paper use the deterministic baseline, $\gamma = 0$. In that deterministic case, stability is equivalent to all eigenvalues of $\boldsymbol{\Phi} - \frac{\rho}{2}\boldsymbol{I}$ having negative real parts. With multiplicative shocks, the same definition requires the discounted second moments induced by the closed-loop process to be finite. This stability requirement corresponds to the standard growth-theory condition that the discount rate be sufficiently large for infinite-horizon utility to be finite.

Assumption 1. *There exists a stabilizing solution* $\left(\boldsymbol{X}^1, \boldsymbol{X}^2, \ldots, \boldsymbol{X}^n\right)$ *of the stacked algebraic Riccati Equation* (11)*.*

Assumption 1 is an existence condition for a stabilizing solution of the coupled Riccati system, not a uniqueness claim. This distinction is standard in infinite-horizon nonzero-sum linear-quadratic differential games, where coupled Riccati equations can have multiple stabilizing solutions or none (Papavassilopoulos et al., 1979; Basar and Olsder, 1999; Engwerda, 2005).

Theorem 1 (Verification of the competitive feedback rule)**.** *Fix a solution* $\{\boldsymbol{X}^j\}_{j=1}^n$ *of Equation* (11)*. Suppose the induced closed-loop process is stabilizing in the sense of Definition 2. For each firm i, take rivals' feedback rules* $x_j(\boldsymbol{z}) = \mu(\boldsymbol{X}_j^j)^T \boldsymbol{z}$ *as given, where* $\boldsymbol{X}_j^j$ *denotes the jth column of* $\boldsymbol{X}^j$*. Then*

$$x_i(\boldsymbol{z}) = \mu(\boldsymbol{X}_i^i)^T \boldsymbol{z}$$

is firm i's optimal Markov response among admissible controls for which the state equation is well defined, the expected discounted quadratic payoff is finite, and the boundary term satisfies

$$\liminf_{T\to\infty} \mathbb{E}_0 \left[e^{-\rho T} V^i(\boldsymbol{z}_T)\right] \geq 0.$$

Therefore, conditional on existence and stabilization, a solution of the stacked Riccati system induces a linear Markov perfect equilibrium in the stated admissible class.

The theorem verifies optimality among all controls in the stated admissible class, not merely within the class of linear feedback rules. Its proof is in Section A.4, and Section C.3 describes the numerical selection rule used in the reported counterfactuals.

We next use the closed-loop technology transition matrix $\mathbf{\Phi}$ to characterize the economy's deterministic balanced growth path.

2.2.4 Deterministic Balanced Growth Path

Set $\gamma = 0$ and consider the closed-loop technology system $\dot{\boldsymbol{z}}_t = \mathbf{\Phi}\boldsymbol{z}_t$. If $\mathbf{\Phi}$ has a simple real dominant eigenvalue $g > 0$ with a strictly positive right eigenvector $\bar{\boldsymbol{z}}$, then $\boldsymbol{z}_t = e^{gt}c\bar{\boldsymbol{z}}$ is a balanced growth path. Thus, the long-run growth rate is the dominant eigenvalue of the technology transition matrix, and the balanced-growth-path cross-section is the associated eigenvector.

Because the static mapping and R&D policies are linear in $\boldsymbol{z}_t$, quantities and R&D grow at rate g along this path. Output, consumption, and profits grow at rate $2g$ because they are quadratic in knowledge capital. The formal spectral condition, proof, growth-rate table, and partial-equilibrium diagram are collected in Section A.6. The quantitative analysis below is evaluated at the observed 2017 knowledge-capital distribution, so this balanced-growth-path characterization is a long-run implication of the solved dynamic system rather than the object used for the main counterfactual comparisons.

2.2.5 Household Welfare

Once the equilibrium is solved, household welfare can be computed for any initial knowledge-capital distribution with minimal additional cost. Let $V(\boldsymbol{z}_0)$ denote household welfare in the competitive equilibrium:

$$V(\boldsymbol{z}_0) = \mathbb{E}_0\left[\int_0^\infty \exp(-\rho t)\left\{Y_t - \sum_i x_{i,t}^2\right\} dt \,\middle|\, \boldsymbol{z}_0\right].$$

Total output is

$$Y_t = \boldsymbol{z}_t^T \boldsymbol{Q} \boldsymbol{z}_t$$

where

$$\boldsymbol{Q} = \boldsymbol{N}^T\left(\frac{\zeta}{L}\boldsymbol{J} + \frac{1}{2}\mathbf{\Sigma} + \boldsymbol{I}\right)\boldsymbol{N}$$

(see Section A.3 for the derivation). The matrix $\boldsymbol{Q}$ is the output-flow matrix, not the sum of the firm payoff matrices: $\sum_i \boldsymbol{Q}_i = \boldsymbol{N}^T\boldsymbol{N}$. Thus, $\boldsymbol{Q} - \boldsymbol{N}^T\boldsymbol{N}$ is the part of static output flow not captured by aggregate gross producer profits, the static counterpart of the appropriability force in the wedge decomposition below. After firms' equilibrium policies are fixed, the remaining valuation equation contains no controls. Because the closed-loop process and the payoff are linear-quadratic, household welfare has the form $V(\boldsymbol{z}) = \boldsymbol{z}^T \boldsymbol{X} \boldsymbol{z}$. Substituting this quadratic form into the valuation equation gives the Lyapunov equation

$$\begin{aligned}0 = \boldsymbol{Q} - \mu^2 \widetilde{\boldsymbol{X}}^T \widetilde{\boldsymbol{X}} + \boldsymbol{X}\left(\boldsymbol{\Phi} - \frac{\rho}{2}\boldsymbol{I}\right) + \left(\boldsymbol{\Phi} - \frac{\rho}{2}\boldsymbol{I}\right)^T \boldsymbol{X} \\ + \gamma^2 \mathcal{D}(\boldsymbol{X})\end{aligned} \tag{12}$$

Therefore, $\boldsymbol{X}$ is obtained as the solution of Equation (12), and welfare in the competitive equilibrium is $V(\boldsymbol{z}_0) = \boldsymbol{z}_0^T \boldsymbol{X} \boldsymbol{z}_0$. The valuation equation and producer-value analogue are derived in Section A.5.

2.2.6 Producer Value

We also characterize aggregate producer value in the competitive equilibrium. This object provides a measure of producers' dynamic value that can be compared with household welfare and with the counterfactual allocations below. Aggregate real profit is

$$\sum_i \pi_{i,t} = \boldsymbol{q}_t^T \boldsymbol{q}_t = \boldsymbol{z}_t^T \boldsymbol{N}^T \boldsymbol{N} \boldsymbol{z}_t.$$

Relative to the household-welfare calculation in Section 2.2.5, aggregate producer value is obtained by solving a modified version of Equation (12) in which $\boldsymbol{Q}$ is replaced with $\boldsymbol{N}^T\boldsymbol{N}$. Denoting the solution by $\boldsymbol{X}^P$, aggregate producer value is $V^P(\boldsymbol{z}_0) = \boldsymbol{z}_0^T \boldsymbol{X}^P \boldsymbol{z}_0$.

2.3 Planner, Monopoly, and Constrained Allocations

We examine counterfactual allocations and compare them with the competitive equilibrium. The purpose of these allocations is to decompose distortions, not to claim that each allocation is directly implementable as a policy. Each counterfactual varies two margins: the static product-market allocation and the dynamic R&D rule. The scenario labels are two-letter codes: the first letter denotes the static product-market allocation, and the second denotes the dynamic R&D rule. The label C denotes competition, S the social planner, and M monopoly. Table 1 summarizes the allocations defined in this section.

We first analyze the allocation chosen by a benevolent social planner who aims to maximize household welfare. We then consider the monopoly case, in which a monopolist determines production and R&D to maximize aggregate producer value. Finally, we define constrained counterfactuals that combine the competitive static product-market allocation with either constrained-planner or constrained-monopolist control of R&D.

Table 1: Counterfactual Allocations in the Model

Scenario	Static mapping	Dynamic R&D controller	Flow payoff before R&D cost
CC	$\boldsymbol{q} = \boldsymbol{N}\boldsymbol{z}$	Firm i maximizes own value	$\boldsymbol{z}^T\boldsymbol{Q}_i\boldsymbol{z}$
CS	$\boldsymbol{q} = \boldsymbol{N}\boldsymbol{z}$	Constrained planner maximizes household welfare	$\boldsymbol{z}^T\boldsymbol{Q}\boldsymbol{z}$
CM	$\boldsymbol{q} = \boldsymbol{N}\boldsymbol{z}$	Monopolist maximizes aggregate producer value	$\boldsymbol{z}^T\boldsymbol{N}^T\boldsymbol{N}\boldsymbol{z}$
CC+s	$\boldsymbol{q} = \boldsymbol{N}\boldsymbol{z}$	Firm i maximizes own value with private R&D cost $(1-s)x_i^2$	$\boldsymbol{z}^T\boldsymbol{Q}_i\boldsymbol{z}$
SS	$\boldsymbol{q} = \boldsymbol{N}^*\boldsymbol{z}$	Social planner maximizes household welfare	$\frac{1}{2}\boldsymbol{z}^T\boldsymbol{N}^*\boldsymbol{z}$
MM	$\boldsymbol{q} = \boldsymbol{N}^M\boldsymbol{z}$	Monopolist maximizes aggregate producer value	$\boldsymbol{z}^T\boldsymbol{P}^M\boldsymbol{z}$

Notes: Each dynamic objective subtracts R&D costs from the flow payoff shown in the last column. In CC+s, firms face the private cost $(1-s)x_i^2$; subsidy payments are lump-sum financed, so household welfare subtracts the full resource cost $\boldsymbol{x}^T\boldsymbol{x}$. The constrained scenarios CS and CM keep the competitive static mapping fixed and change only the dynamic R&D controller. CM is an R&D-only common-control case: product-market quantities remain competitive while R&D is chosen to maximize aggregate producer value.

2.3.1 Social Planner

To analyze the socially optimal allocation of R&D and its implications for observed-state output growth and household welfare, we formulate the social planner's problem. This full social-planner allocation corresponds to scenario SS in Table 1. The social planner aims to maximize the representative household's lifetime utility, subject to technological constraints. As in the competitive equilibrium, the social planner's problem can be separated into a static problem and a dynamic problem.

Static Problem First, we characterize the static allocation in the social planner's problem. The social planner maximizes final-good output given the vector of knowledge capital $\boldsymbol{z}_t$ in each period:

$$\max_{\boldsymbol{a}_t,\boldsymbol{b}_t,\boldsymbol{l}_t,\boldsymbol{q}_t} Y_t = \boldsymbol{q}_t^T\boldsymbol{b}_t - \frac{1}{2}\boldsymbol{q}_t^T\boldsymbol{\Sigma}\boldsymbol{q}_t$$

subject to

$$\begin{aligned} q_{i,t} &= a_{i,t}l_{i,t} \quad i = \{1,2,\ldots,n\} \\ z_{i,t} &= \zeta a_{i,t} + b_{i,t} \quad i = \{1,2,\ldots,n\} \\ L &= \sum_i l_{i,t} \end{aligned} \tag{13}$$

The first constraint represents the production technology of each firm, the second governs the allocation of each firm's knowledge capital between labor productivity and product quality, and the third ensures the labor market clears. Solving the planner's static problem yields quantities as linear functions of knowledge capital:

$$\boldsymbol{q}_t^* = \boldsymbol{N}^* \boldsymbol{z}_t$$

and a quadratic representation of output in knowledge capital

$$Y_t^* = \frac{1}{2} \boldsymbol{z}_t^T \boldsymbol{N}^* \boldsymbol{z}_t$$

where

$$\boldsymbol{N}^* = \left(2\frac{\zeta}{L} \boldsymbol{J} + \boldsymbol{\Sigma} \right)^{-1}$$

(see Section B.1).

Dynamic Problem Next, we characterize the dynamic allocation in the social planner's problem. Given the static planner output coefficient, the planner chooses R&D to maximize

$$V^{SS}(\boldsymbol{z}_0) \equiv \max_{\{x_t\}_{t \geq 0}} \mathbb{E}_0 \left[\int_0^\infty \exp(-\rho t) \left\{ \frac{1}{2} \boldsymbol{z}_t^T \boldsymbol{N}^* \boldsymbol{z}_t - \boldsymbol{x}_t^T \boldsymbol{x}_t \right\} dt \right],$$

subject to the knowledge accumulation law in Equation (4). This is again a linear-quadratic control problem, so the optimal R&D rule is linear in knowledge capital:

$$\boldsymbol{x} = \mu \boldsymbol{X}^{SS} \boldsymbol{z} \tag{14}$$

and the induced law of motion can be written as

$$d\boldsymbol{z}_t = \boldsymbol{\Phi}^{SS} \boldsymbol{z}_t dt + \gamma \operatorname{diag}(\boldsymbol{z}_t) d\boldsymbol{W}_t$$

where

$$\boldsymbol{\Phi}^{SS} \equiv \boldsymbol{\Omega} - \delta \boldsymbol{I} + \mu^2 \boldsymbol{X}^{SS}$$

Substituting the quadratic planner value and the linear R&D rule into the planner's HJB yields a single algebraic Riccati equation:

$$0 = \frac{1}{2} \boldsymbol{N}^* - \mu^2 \left(\boldsymbol{X}^{SS} \right)^2 + \boldsymbol{X}^{SS} \left(\boldsymbol{\Phi}^{SS} - \frac{\rho}{2} \boldsymbol{I} \right) + \left(\boldsymbol{\Phi}^{SS} - \frac{\rho}{2} \boldsymbol{I} \right)^T \boldsymbol{X}^{SS}$$

$$+\gamma^2 \mathcal{D}\left(\boldsymbol{X}^{SS}\right) \tag{15}$$

Therefore, $\boldsymbol{X}^{SS}$ is obtained as the solution of Equation (15), and household welfare under the optimal allocation is $V^{SS}(\boldsymbol{z}_0) = \boldsymbol{z}_0^T \boldsymbol{X}^{SS} \boldsymbol{z}_0$. The HJB and derivative steps are in Section B.2. Unlike the competitive equilibrium, which requires one Riccati equation per firm, the social-planner problem requires only a single Riccati equation.

2.3.2 Monopoly

As a second counterfactual scenario, we consider the monopoly case, in which a single monopolist controls both product-market production and R&D. This full monopoly allocation corresponds to scenario MM. The monopolist determines all production and R&D effort to maximize aggregate producer value. In the static product-market block, the monopolist chooses $\boldsymbol{a}_t$, $\boldsymbol{b}_t$, and $\boldsymbol{q}_t$ to maximize aggregate real profit

$$\Pi_t \equiv \frac{\boldsymbol{q}_t^T \boldsymbol{p}_t}{P_t} - \frac{w_t}{P_t} \sum_i l_{i,t}.$$

Solving the monopolist's static problem yields

$$\boldsymbol{q}_t^M = \boldsymbol{N}^M \boldsymbol{z}_t, \qquad \boldsymbol{N}^M \equiv \frac{1}{2}\left(\frac{\zeta}{L}\boldsymbol{J} + \boldsymbol{\Sigma}\right)^{-1},$$

and aggregate firm profit can be written as

$$\Pi_t^M = \boldsymbol{z}_t^T \boldsymbol{P}^M \boldsymbol{z}_t, \qquad \boldsymbol{P}^M \equiv (\boldsymbol{N}^M)^T \boldsymbol{\Sigma} \boldsymbol{N}^M.$$

The superscript P denotes producer value, distinguishing the monopolist's objective from household welfare under the induced policy. The monopolist's dynamic producer-value problem is therefore

$$V^{P,MM}(\boldsymbol{z}_0) \equiv \max_{\{\boldsymbol{x}_t\}_{t\geq 0}} \mathbb{E}_0 \left[\int_0^\infty \exp\left(-\rho t\right) \left\{\boldsymbol{z}_t^T \boldsymbol{P}^M \boldsymbol{z}_t - \boldsymbol{x}_t^T \boldsymbol{x}_t\right\} dt\right],$$

subject to the knowledge accumulation law. This problem is solved by replacing $\boldsymbol{N}^*/2$ with $\boldsymbol{P}^M$ in Equation (15). Let $\boldsymbol{X}^{P,MM}$ denote the solution. The monopoly R&D policy is then $\boldsymbol{x}_t = \mu \boldsymbol{X}^{P,MM} \boldsymbol{z}_t$. Because the monopolist maximizes producer value rather than household welfare, the household welfare reported for MM is computed separately from a Lyapunov equation using the monopoly output matrix $\boldsymbol{Q}^M$ and the technology transition induced by

$\boldsymbol{X}^{P,MM}$. For the derivation of $\boldsymbol{N}^M$, $\boldsymbol{P}^M$, $\boldsymbol{Q}^M$, and the welfare equation, see Section B.3.

2.3.3 Separating Product-Market and R&D Control

We also consider counterfactual scenarios in which control of production differs from control of R&D. These constrained counterfactuals keep the static competitive mapping $\boldsymbol{q}_t = \boldsymbol{N}\boldsymbol{z}_t$ fixed and change only the dynamic R&D controller. They are useful because they separate innovation incentives from contemporaneous product-market reallocation.

First, consider scenario CS, in which firms competitively determine production quantities, and given this static competitive equilibrium, a benevolent constrained planner then selects the optimal allocation of R&D. Starting from an initial knowledge-capital distribution $\boldsymbol{z}_0$, the constrained planner maximizes the value

$$V^{CS}(\boldsymbol{z}_0) \equiv \max_{\{\boldsymbol{x}_t\}_{t\geq 0}} \mathbb{E}_0 \left[\int_0^\infty \exp(-\rho t) \left\{ \boldsymbol{z}_t^T \boldsymbol{Q} \boldsymbol{z}_t - \boldsymbol{x}_t^T \boldsymbol{x}_t \right\} dt \right],$$

subject to

$$d\boldsymbol{z}_t = \left((\boldsymbol{\Omega} - \delta \boldsymbol{I}) \boldsymbol{z}_t + \mu \boldsymbol{x}_t \right) dt + \gamma \operatorname{diag}(\boldsymbol{z}_t) d\boldsymbol{W}_t.$$

Unlike in Section 2.3.1, the constrained planner's gross instantaneous return is $\boldsymbol{z}_t^T \boldsymbol{Q} \boldsymbol{z}_t$ rather than $\frac{1}{2}\boldsymbol{z}_t^T \boldsymbol{N}^* \boldsymbol{z}_t$. Therefore, we obtain the constrained planner's allocation by solving a modified version of Equation (15), in which $\boldsymbol{N}^*/2$ is replaced with $\boldsymbol{Q}$. Denoting the solution of the modified Riccati equation by $\boldsymbol{X}^{CS}$, household welfare under the constrained-planner allocation is given by $V^{CS}(\boldsymbol{z}_0) = \boldsymbol{z}_0^T \boldsymbol{X}^{CS} \boldsymbol{z}_0$.

The constrained monopolist in scenario CM is defined analogously, except that the dynamic controller maximizes aggregate producer value rather than household welfare. Given the competitive static product-market allocation, aggregate real profit is $\boldsymbol{z}_t^T \boldsymbol{N}^T \boldsymbol{N} \boldsymbol{z}_t$. Hence the constrained monopolist chooses R&D to maximize

$$V^{P,CM}(\boldsymbol{z}_0) \equiv \max_{\{\boldsymbol{x}_t\}_{t\geq 0}} \mathbb{E}_0 \left[\int_0^\infty \exp(-\rho t) \left\{ \boldsymbol{z}_t^T \boldsymbol{N}^T \boldsymbol{N} \boldsymbol{z}_t - \boldsymbol{x}_t^T \boldsymbol{x}_t \right\} dt \right],$$

subject to the same knowledge accumulation law. This allocation is obtained by replacing $\boldsymbol{N}^*/2$ with $\boldsymbol{N}^T\boldsymbol{N}$ in Equation (15). Let $\boldsymbol{X}^{P,CM}$ denote the resulting producer-value coefficient, so the induced R&D rule is $\boldsymbol{x}_t = \mu \boldsymbol{X}^{P,CM} \boldsymbol{z}_t$. Household welfare under this constrained-monopolist R&D policy is then computed from the same Lyapunov equation used in Section 2.2.5, with the competitive output matrix $\boldsymbol{Q}$ and the technology transition induced by $\boldsymbol{X}^{P,CM}$. Denoting the resulting welfare coefficient by $\boldsymbol{X}^{CM}$, household welfare is $V^{CM}(\boldsymbol{z}_0) = \boldsymbol{z}_0^T \boldsymbol{X}^{CM} \boldsymbol{z}_0$.

Table 2: Data Requirements for Sample Construction

Requirement	Role in the empirical implementation
Compustat coverage	Provides accounting variables used to measure profits, sales, costs, and R&D.
Hoberg-Phillips product-similarity data	Provides the empirical product-similarity matrix S used to construct $\mathbf{\Sigma}$.
DISCERN-linked patent records	Links public firms to patents for technology portfolios.
Positive gross profits	Ensures that $q_i = \sqrt{\pi_i}$ is well defined under $\pi_i = q_i^2$.
At least ten patents in a five-year window	Ensures nontrivial linked patent portfolios for measuring technology shares.

In the numerical analysis, we also compare these planner and monopoly allocations with the uniform-subsidy equilibrium CC+s.

3 Data and Estimation

We map the model's product-market rivalry and technology-spillover networks, knowledge capital, and dynamic parameters to observed data. We use a 1989–2019 panel to construct annual networks, recover firm-level knowledge capital, and estimate technology spillovers. The quantitative baseline and counterfactuals are solved at the observed 2017 cross section of 757 firms.

3.1 Sample Construction and Empirical Mapping

For each year, we keep firms that satisfy the data requirements in Table 2. The resulting annual samples range from 418 firms in 1989 to 802 firms in 1998, and the 2017 quantitative baseline contains 757 firms. The 2017 baseline captures most activity in the relevant patenting-public-firm universe: among 2017 Compustat firms with positive gross profits and at least one DISCERN-linked patent in the 2015–2019 grant window, the baseline firms account for about 91% of sales, 98% of observed R&D, and 94% of gross profits. Appendix Section C.1 and Table 15 report the corresponding patent coverage and raw network-overlap measures. Table 5 summarizes the estimated parameters and their sources.

Table 3 summarizes the empirical procedure: selecting the sample, constructing the two networks, recovering firm-level quantities and knowledge capital from Compustat variables, and estimating or calibrating the remaining dynamic parameters.

Table 3: Mapping Data to Model Objects

Model object	Data source or empirical object	Mapping to model
$\boldsymbol{S}, \boldsymbol{\Sigma}(\alpha)$	Hoberg-Phillips product similarity	Hoberg cosine similarities measure $\boldsymbol{S}$; the solver uses $\boldsymbol{\Sigma} = \alpha \boldsymbol{S} + (1-\alpha)\boldsymbol{I}$, with diagonal entries normalized to one.
$\tilde{\boldsymbol{\Omega}}$	DISCERN patents and CPC groups	Jaffe overlaps are computed from five-year patent shares; the diagonal is set to zero and rows are normalized by recipient.
$\boldsymbol{q}$	Gross profits, revt − cogs	The static first-order condition implies $\pi_i = q_i^2$, so quantities are recovered as $q_i = \sqrt{\pi_i}$.
ζ/L	Aggregate COGS and recovered quantities	The aggregate production-cost condition identifies the ratio ζ/L.
$\boldsymbol{z}$	Recovered $\boldsymbol{q}$, $\boldsymbol{\Sigma}$, and ζ/L	The static equilibrium condition recovers $\boldsymbol{z}_t = (2(\zeta/L)\boldsymbol{J} + \boldsymbol{\Sigma} + \boldsymbol{I})\boldsymbol{q}_t$.
β	Panel growth of recovered knowledge capital and R&D controls	Spillover intensity is estimated from the knowledge-capital law of motion; observed R&D enters only as a regression control.
μ and δ	R&D expenditure and observed-state output growth	Own-R&D productivity and depreciation are jointly calibrated in the 2017 baseline; R&D effort is endogenous in the model.

Notes: Nominal accounting variables are deflated using the GDP deflator. The effective spillover matrix used in the law of motion is $\boldsymbol{\Omega} = \beta\tilde{\boldsymbol{\Omega}}$. R&D effort is endogenous in the model; observed R&D expenditures are used for the law-of-motion regression and calibration moments. The recovered 2017 knowledge-capital vector is used as the initial state in the numerical analysis.

3.2 Product-Market Proximity and Substitutability

Following Pellegrino (2025), we measure the product-similarity matrix $\boldsymbol{S}$ using the Hoberg-Phillips 10-K text similarity data (Hoberg and Phillips, 2016). These cosine-similarity scores predict competitive relationships and provide the empirical counterpart of $\boldsymbol{\Psi}^T\boldsymbol{\Psi}$ in the GHL microfoundation in Section A.1. The numerical analysis constructs the product-rivalry matrix $\boldsymbol{\Sigma}(\alpha) = \alpha\boldsymbol{S} + (1-\alpha)\boldsymbol{I}$ before solving the static and dynamic blocks, so for $i \neq j$, $\sigma_{ij} = \alpha S_{ij}$, while the model normalization imposes $S_{ii} = \sigma_{ii} = 1$.

We take α, which governs the degree of horizontal differentiation across goods, from Pellegrino (2025). That paper disciplines α using the inverse cross-price elasticity reported by Nevo (2001) and shows that the implied price elasticities are aligned with estimates for automobiles, ready-to-eat cereal, and computers (Berry et al., 1995; Nevo, 2001; Goeree, 2008).

Appendix Table 13 reports a local sensitivity exercise that recovers the 2017 knowledge-capital state and recalibrates the dynamic parameters at nearby values of α.

3.3 Technological Proximity

For technological proximity, we use patent data and follow Bloom et al. (2013) and Lucking et al. (2019) to construct measures of technological proximity between firms. Specifically, we use the DISCERN 2 dataset (Arora et al., 2024), which links data on U.S. publicly listed firms from the Compustat database to their patents. Following Bloom et al. (2013), we calculate technological proximity between firms using the Jaffe measure based on Cooperative Patent Classification groups. Patent shares are computed using patents granted from $t-2$ through $t+2$ around each sample year t. We use this centered grant-year window because patent grants are dated with delay relative to the underlying inventive activity: in the DISCERN-linked patents with filing years observed, the median lag between filing year and grant year is three years, and the mean lag is 3.09 years. The window therefore smooths sparse annual patent classifications and partially offsets grant-date delay, yielding a measure of firms' persistent technological locations.

Let T_i denote the vector of firm i's patent shares across technology classes. The raw Jaffe overlap (Jaffe, 1986) between firms i and j is

$$\hat{\omega}_{ij} = \frac{T_i T_j'}{\left(T_i T_i'\right)^{1/2}\left(T_j T_j'\right)^{1/2}}.$$

We measure T_i using Cooperative Patent Classification groups and construct firm-pair overlaps from the normalized technology vectors.

As in Liu and Ma (2026), we convert these raw overlaps into recipient-normalized spillover weights. We set the diagonal entries to zero and normalize each recipient firm's row:

$$\tilde{\omega}_{ij} = \frac{\hat{\omega}_{ij}}{\sum_{k\neq i}\hat{\omega}_{ik}}, \qquad j \neq i.$$

Rows with no off-diagonal overlap are left at zero. This normalization makes β capture the common intensity of spillovers received by a firm, while $\tilde{\omega}_{ij}$ determines how that exposure is allocated across source firms. Thus, for firms with positive technology overlap, the number of technology neighbors changes the composition of received spillover exposure but not the row total before scaling by β. Although the raw Jaffe overlap is symmetric, recipient normalization generally makes $\tilde{\mathbf{\Omega}}$ asymmetric. This is the empirical counterpart of the model's row-receiver convention: $\tilde{\omega}_{ij}$ is the share of firm i's spillover exposure accounted for by source firm j, and the effective spillover matrix used in the law of motion is $\mathbf{\Omega} = \beta\tilde{\mathbf{\Omega}}$.

The product-market and technology networks are empirically distinct. In the 2017 baseline, the firm-level Spearman correlation between product-market centrality and technology-source centrality is 0.22, or 0.27 using sales-weighted destination centralities.

3.4 Knowledge Capital Distribution

Next, we recover firm-level quantities, the ratio ζ/L, and knowledge capital $\boldsymbol{z}_t$ from Compustat accounting variables. The recovery proceeds in three steps.

First, we measure firm-level gross profit $\pi_{i,t}$ as total revenue (revt) minus cost of goods sold (cogs), with nominal variables deflated using the GDP deflator. In the model, the static Cournot first-order condition implies $\pi_{i,t} = q_{i,t}^2$. This identity is the reason we require positive gross profits in the sample: it lets us recover quantities as $q_{i,t} = \sqrt{\pi_{i,t}}$.

Second, we identify ζ/L from aggregate production costs. We interpret Compustat cost of goods sold as the empirical counterpart of variable production costs in the static block. Under this interpretation, the model implies

$$\frac{\zeta}{L}\boldsymbol{q}_t^T \boldsymbol{J} \boldsymbol{q}_t = \text{total production cost at time } t.$$

We calculate total production cost as the sum of cogs for all firms in the sample. Since only the ratio ζ/L enters the static equilibrium, ζ and L cannot be separately identified.

Third, we recover the knowledge-capital vector from the static equilibrium condition. Given the recovered quantities and product-rivalry matrix, Equation (7) implies

$$\boldsymbol{z}_t = \left(2\frac{\zeta}{L}\boldsymbol{J} + \boldsymbol{\Sigma} + \boldsymbol{I}\right)\boldsymbol{q}_t.$$

Because $\boldsymbol{\Sigma}$ has unit diagonal in the model, the diagonal of this recovery equation contains two identity terms: one from the unit diagonal of $\boldsymbol{\Sigma}$ and one from the Cournot own-output term in the static first-order condition. Equivalently, the empirical implementation adds $2\boldsymbol{I}$ to the stored off-diagonal product-proximity matrix. The recovered $\boldsymbol{z}_t$ is then used to estimate the law of motion and, for 2017, as the initial distribution in the numerical analysis.

3.5 Knowledge-Capital Law of Motion

In the empirical implementation, $x_{i,t}$ denotes R&D effort, measured as the square root of Compustat R&D expenditure (xrd), so that $x_{i,t}^2$ corresponds to observed R&D expenditure, consistent with the model's resource constraint. We estimate the spillover intensity β from panel variation in recovered knowledge capital. Dividing the drift of the model's

Table 4: Knowledge-Capital Growth and Technology Exposure

	(1)	(2)	(3)	(4)
Technology exposure, $\Omega z_j/z_i$	0.039***	0.021***	0.022**	0.025***
	(0.008)	(0.008)	(0.009)	(0.008)
Product-market exposure, $\Sigma z_j/z_i$		-0.008***	-0.010***	-0.009***
		(0.001)	(0.001)	(0.001)
Controls: x_i/z_i, $\log z_i$	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes
Year/interacted FE	Year	Year	NAICS4-year	Tech-year
Observations	18,724	18,724	18,724	18,724

Notes: The dependent variable is $\Delta \log z_i$. Standard errors, clustered by NAICS4-by-year, are in parentheses. All columns use the full sample, OLS, exact-calendar growth, firm fixed effects, x_i/z_i, and $\log z_i$. Variables are cleaned by 0.5 percent winsorization before and after exposure construction, with $\log z_i$ trimmed. Product-market exposure is included where reported. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

knowledge law of motion by $z_{i,t}$ shows that spillovers affect proportional knowledge growth through $\sum_{j \neq i} \tilde{\omega}_{ij,t} z_{j,t}/z_{i,t}$, while own R&D enters through $x_{i,t}/z_{i,t}$ and depreciation enters as a common term. The depreciation component is absorbed by fixed effects and residual common drift in the panel specification, giving the annual-difference analogue:

$$\log z_{i,t+1} - \log z_{i,t} = \beta \sum_{j \neq i} \tilde{\omega}_{ij,t} \frac{z_{j,t}}{z_{i,t}} + \text{Controls}_{i,t} + \epsilon_{i,t} \tag{16}$$

where $z_{i,t}$ denotes firm i's knowledge capital at time t, and $\tilde{\omega}_{ij,t}$ measures the recipient-normalized technological proximity between firms i and j at time t. The controls include year and firm fixed effects, $\log z_{i,t}$, and R&D-effort intensity $x_{i,t}/z_{i,t}$. Own R&D is included as a conditioning variable, not as the source of structural identification for μ; we calibrate μ and δ below using aggregate moments in the 2017 baseline. We also report specifications that add product-market-neighbor exposure to separate technology-neighbor exposure from product-market-neighbor exposure.

Table 4 reports estimates of Equation (16). Column (1) reports the raw technology-exposure association: the coefficient on $\sum_{j \neq i} \tilde{\omega}_{ij,t} z_{j,t}/z_{i,t}$ is 0.039 and statistically significant. Column (2) adds product-market-neighbor exposure, $\sum_{j \neq i} s_{ij,t} z_{j,t}/z_{i,t}$, to separate technology exposure from product-market-neighbor exposure. The technology coefficient falls to 0.021 but remains positive and statistically significant, while the product-market exposure coefficient is negative. Columns (3) and (4) replace year fixed effects with NAICS4-by-year and modal-technology-section-by-year fixed effects; the corresponding technology coefficients are 0.022 and 0.025. We therefore use the rounded value $\beta = 0.02$

as the central numerical anchor in the quantitative model, with $\beta = 0.01$ and $\beta = 0.03$ as lower and upper OLS sensitivity cases.

3.6 Remaining Dynamic Parameters

Having estimated the spillover intensity β, we set the discount rate ρ externally and use aggregate moments to discipline the remaining dynamic parameters, the R&D coefficient μ and the depreciation rate δ. The baseline sets $\rho = 10\%$ and uses the rounded central spillover intensity $\beta = 0.02$.

We estimate μ and δ jointly in the 2017 competitive-equilibrium baseline by matching two aggregate moments: R&D intensity of 2.6% and average real GDP per capita growth of 1.5% over 1989–2019. For any candidate pair (μ, δ), we solve the 2017 competitive equilibrium at the observed knowledge-capital vector and compute the model-implied R&D intensity and observed-state growth rate, $g_Y \equiv d \log Y_t / dt$. This observed-state growth moment is not the balanced-growth-path eigenvalue characterized in the model section. We measure model-implied R&D intensity as aggregate R&D expenditure, $\sum_i x_i^2$, divided by labor payments plus gross operating profits. The calibration chooses μ and δ to match these two targets jointly, yielding $\mu = 0.0525$ and $\delta = 0.0113$.

Because ρ governs both discounted welfare and the finite-value condition, Appendix Table 14 reports a sensitivity exercise that recalibrates μ and δ at $\rho \in \{5\%, 7.5\%, 10\%\}$. Lower discount rates raise the welfare value of constrained-planner R&D gains but reduce the stability margin, with the $\rho = 5\%$ case close to the finite-value boundary. Thus the baseline $\rho = 10\%$ is conservative for fixed-product-market R&D gains and leaves a wider stability margin across reported scenarios.

Appendix Table 12 propagates the rounded beta grid through the full structural calculation. The main numerical results below use $\beta = 0.02$, with $\beta = 0.01$ and $\beta = 0.03$ as lower and upper sensitivity cases.

3.7 Model Fit and Validation

We evaluate the model's fit using both targeted and non-targeted moments. Figure 1 compares firm-level profits, sales, and R&D expenditures in the model and in the data. Profits are targeted by construction because we recover quantities from the profit identity implied by the static equilibrium. The more informative checks are therefore sales and R&D. The model reproduces the cross-sectional distribution of sales closely, with a correlation of 0.96 between model-implied and observed log sales. It also captures a substantial part of

Table 5: Parameters

Notation	Description	Value	Source
$\boldsymbol{S}$	Product similarity	–	Hoberg and Phillips (2016)
$\tilde{\boldsymbol{\Omega}}$	Technology-spillover exposure weights	–	DISCERN patents, CPC groups
α	Product similarity to rivalry	0.12	Pellegrino (2025)
β	Spillover-intensity scale	0.020	Rounded central estimate from the law of motion
γ	Std. dev. of idiosyncratic shocks	0.00	Deterministic economy in the baseline
ζ/L	Labor augmentation efficiency	0.0040	Compustat, cost of goods sold
ρ	Discount rate	0.10	Baseline choice
μ	R&D coefficient	0.0525	Jointly match R&D share and growth
δ	Depreciation rate	0.0113	Jointly match R&D share and growth

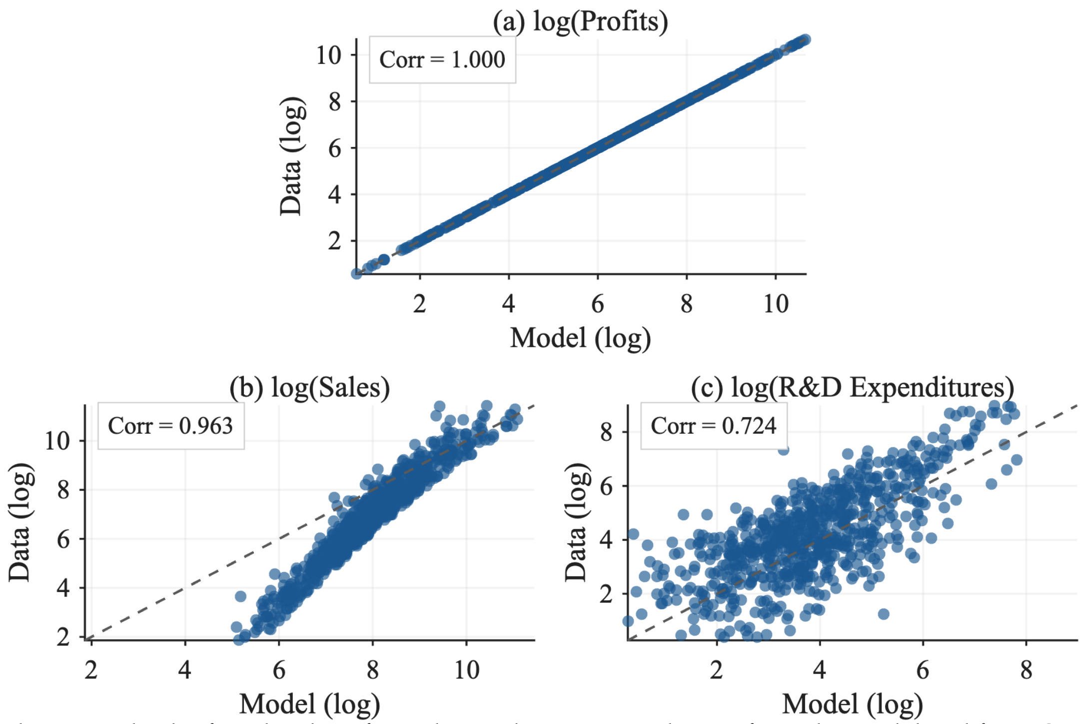
Notes: Dashes indicate matrix-valued empirical objects rather than scalar parameter estimates. The baseline sets $\gamma = 0$ for the deterministic economy. The ratio ζ/L is identified from aggregate production costs; ζ and L are not separately identified. The parameters μ and δ are jointly calibrated in the 2017 competitive baseline to match aggregate R&D intensity and observed-state g_Y. The depreciation rate applies to recovered knowledge capital z, not to an accounting physical or patent-based R&D stock.

the dispersion in innovation activity: the correlation between model-implied and observed log R&D expenditures is 0.72.

As a separate external validation, we construct a depreciated patent stock from DISCERN-linked grants, unused in the recovery or calibration. Log recovered knowledge capital is strongly correlated with the log depreciated patent stock: the overall correlation is 0.65 for count-weighted patents and 0.64 for log-citation-weighted patents, with mean annual correlations of 0.69 and 0.68, respectively.

Figure 2 compares firm-level growth rates of knowledge capital in the model and in the data. This is a stricter validation exercise because firm-level knowledge growth is not matched directly in the calibration of the static block. Although firms grow at a common rate on the deterministic balanced growth path, this validation exercise is not evaluated at that asymptotic object. It computes model-implied local growth rates at the observed 2010 knowledge-capital distribution and networks, so cross-sectional growth heterogeneity reflects transitional dynamics from the empirical state. The data-based growth rate is computed from changes between 2010 and 2017 using the 2010 networks and initial knowledge-capital distribution. The correlation between model-implied and data-based growth rates is 0.579, indicating that the estimated technology-spillover network and R&D incentives generate meaningful cross-sectional variation in subsequent knowledge growth,

Figure 1: Firm-Level Profits, Sales, and R&D Expenditures: Model vs. Data



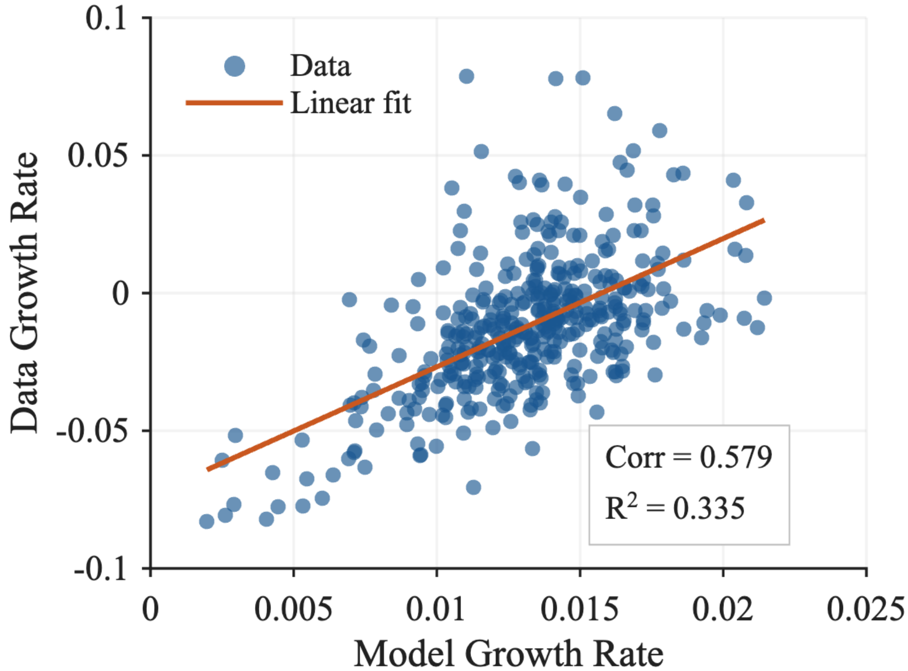
Notes: These panels plot firm-level profits, sales, and R&D expenditures from the model and from Compustat. Profits in the data are measured as "Revenue – Total" (revt) minus "Cost of Goods Sold" (cogs) in Compustat. The model is calibrated so that firm-level profits match those in the data.

even though the model is not designed to match every firm-level growth realization.

4 Numerical Analysis

This section reports the quantitative implications of the estimated model. The main result is that competitive R&D is too low in aggregate and misallocated across firms, but these distortions operate on distinct margins: a welfare-maximizing uniform subsidy captures most of the constrained planner's aggregate R&D-control gain, whereas firm-level wedge heterogeneity and product-market reallocation remain separate margins. We first summarize the computational setup. We then examine the baseline allocation by summarizing social-private R&D wedges and decomposing their components. Next, we use counterfactual allocations to distinguish the effects of dynamic R&D control from those of static product-market reallocation. Finally, we evaluate a uniform R&D subsidy and compare it with the constrained planner allocation.

Figure 2: Knowledge-Capital Growth Rate Comparison: Model vs. Data



Notes: This figure compares firm-level growth rates of knowledge capital implied by the model and computed from the data. The data-based growth rate is measured between 2010 and 2017 and is constructed using the 2010 networks and the initial level of knowledge capital.

4.1 Computation and Scale

The numerical implementation follows the model's separation between static product-market mappings and dynamic R&D rules. For each static allocation, we construct the closed-form matrices that map knowledge capital into quantities, output, and gross producer profit. Conditional on a static block, we solve the relevant dynamic R&D problem and use the resulting feedback rule and value matrices for producer-value and household-welfare calculations.

The counterfactuals below are evaluated at the observed 2017 knowledge-capital distribution. They are therefore comparisons of household welfare and observed-state g_Y starting from the same empirical state, not comparisons of asymptotic balanced-growth-path distributions. The main scenario table should be read as an unconstrained interior benchmark. The full static reallocations use closed-form interior mappings; imposing quantity non-negativity would define a different constrained static problem and cannot be implemented by truncating quantities ex post. Sections C, C.4 and C.5 report observed-state checks for negative interior quantities and R&D efforts, transition-path sign checks, projected-positive welfare robustness, and BGP sign checks. Projecting negative R&D effort to zero changes observed-state g_Y by less than 0.003 percentage points in every scenario and changes welfare by at most 0.13 percentage points.

The competitive dynamic problem solves a coupled system of Riccati equations for all firms' quadratic value functions. The implementation uses blockwise dense matrix updates, which lets the 2017 baseline solve with 757 firms while preserving a continuous state space and the full empirical product-market and technology-spillover networks. The monopoly and planner dynamic problems each reduce to a single Riccati equation because one decision maker internalizes all R&D choices. Household welfare and aggregate producer value are then computed from continuous-time Lyapunov equations. Sections C.2 and C.3 and Table 16 give the update equation, selection rule, numerical checks, and computational scale.

4.2 Social-Private R&D Wedges

Before turning to counterfactual allocations, we first compare private and social innovation incentives in the competitive baseline allocation. We compare private and social marginal returns to R&D at the observed 2017 knowledge-capital vector. The private marginal return is the marginal increase in firm value induced by one additional unit of firm i's R&D effort before subtracting the marginal cost of that effort. Since one additional unit of x_i raises the drift of z_i by μ, this private marginal benefit is

$$\mathrm{PMR}_i(\boldsymbol{z}) \equiv \mu \frac{\partial V^i(\boldsymbol{z})}{\partial z_i} = 2\mu \left(\boldsymbol{X}_i^i\right)^\top \boldsymbol{z}.$$

In vector form, $\mathrm{PMR}(\boldsymbol{z}) = 2\mu \widetilde{\boldsymbol{X}} \boldsymbol{z}$. Similarly, define the social marginal return as the increase in household welfare from one additional unit of firm i's R&D effort,

$$\mathrm{SMR}_i(\boldsymbol{z}) \equiv \mu \frac{\partial V(\boldsymbol{z})}{\partial z_i} = 2\mu \, \boldsymbol{e}_i^\top \boldsymbol{X} \boldsymbol{z}.$$

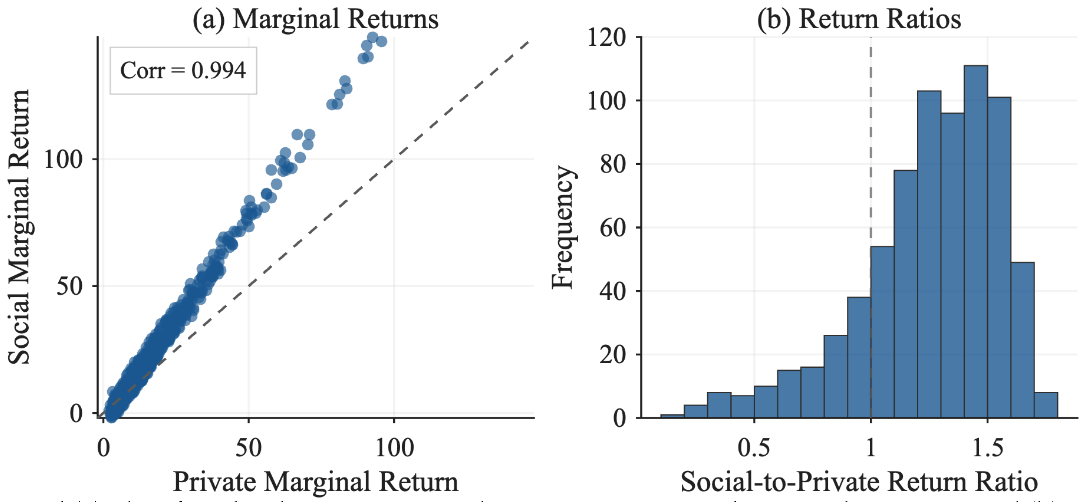
In vector form, $\mathrm{SMR}(\boldsymbol{z}) = 2\mu \boldsymbol{X} \boldsymbol{z}$. The social-private R&D wedge and the social-to-private return ratio are

$$\Delta_i(\boldsymbol{z}) \equiv \mathrm{SMR}_i(\boldsymbol{z}) - \mathrm{PMR}_i(\boldsymbol{z}), \qquad \mathcal{W}_i(\boldsymbol{z}) \equiv \frac{\mathrm{SMR}_i(\boldsymbol{z})}{\mathrm{PMR}_i(\boldsymbol{z})}.$$

When both marginal returns are positive, we also report the local FOC-subsidy measure

$$s_i^{\mathrm{loc}}(\boldsymbol{z}) \equiv 1 - \frac{\mathrm{PMR}_i(\boldsymbol{z})}{\mathrm{SMR}_i(\boldsymbol{z})}.$$

Figure 3: Social-Private R&D Wedges: Marginal Returns and Return Ratios



Notes: Panel (a) plots firm-level private marginal returns against social marginal returns. Panel (b) plots the distribution of $\mathrm{SMR}_i(z)/\mathrm{PMR}_i(z)$ among firms with $\mathrm{PMR}_i(z) > 0$ and $\mathrm{SMR}_i(z) > 0$; the distribution trims the top and bottom 2 percent for readability, and the dashed vertical line marks equality of social and private returns. Returns are computed using the estimated networks and firm-level knowledge capital in 2017.

Table 6: Social-Private R&D Wedge at the Observed 2017 State

Scenario	SMR>PMR (%)	Ratio>1.5 (%)	Ratio>2 (%)	Median ratio	Median local FOC subsidy (%)
CC	82.1	22.3	0.1	1.30	23.3

Notes: $\mathrm{PMR}_i(z) = 2\mu(X_i^i)^T z$ and $\mathrm{SMR}_i(z) = 2\mu e_i^T X z$. Ratios and local FOC subsidies use the sample with $\mathrm{PMR}_i(z) > 0$ and $\mathrm{SMR}_i(z) > 0$, which contains 739 of 757 firms in CC. The local FOC subsidy is $1 - \mathrm{PMR}_i(z)/\mathrm{SMR}_i(z)$ and is an observed-state wedge measure, not a solved firm-specific policy.

This object is an observed-state first-order-condition measure, not a solved firm-specific subsidy or targeted-policy counterfactual; we use it only to summarize the sign and magnitude of the local wedge.

Figure 3 summarizes the firm-level wedge measures by plotting marginal-return levels and the social-to-private return-ratio distribution. The model-implied correlation between private and social marginal returns is high (0.994 after the figure's percentile trimming), but social returns tend to exceed private returns, especially among firms with high R&D returns.

The wedge distribution previews the uniform-subsidy result below. The fact that social returns exceed private returns for many firms is consistent with aggregate R&D underprovision, while the dispersion indicates that the underprovision is heterogeneous across firms. Table 6 establishes the size and dispersion of the wedge. It is computed on the 739 of 757 firms with positive private and social marginal returns. In this sample, the

Table 7: Social-Private R&D Wedge Components by Return-Ratio Decile

Decile	Median SMR/PMR	Mean wedge	APP	RP	RRC
1	0.58	-2.29	11.47	-14.63	0.86
2	0.94	-0.50	13.34	-14.71	0.87
3	1.10	1.02	14.84	-14.68	0.86
4	1.19	2.40	16.42	-14.89	0.87
5	1.27	3.84	17.82	-14.83	0.85
6	1.35	5.30	18.91	-14.43	0.82
7	1.42	8.61	22.50	-14.70	0.82
8	1.48	14.60	28.38	-14.58	0.80
9	1.54	23.30	36.58	-13.99	0.71
10	1.66	17.85	30.51	-13.30	0.64

Notes: Deciles sort firms by $\log(\mathrm{SMR}_i(z)/\mathrm{PMR}_i(z))$ in the 739-firm positive marginal-return sample at the observed 2017 state. The table averages exact deterministic Lyapunov components defined in Section B.5. Mean wedge is $\mathrm{SMR}_i(z) - \mathrm{PMR}_i(z)$. APP, RP, and RRC denote appropriability, rival-profit, and rival R&D resource-cost components. The decomposition is an accounting exercise, not a targeted-policy counterfactual.

median social-to-private return ratio is 1.30, and social returns exceed private returns for 82.1% of firms. The dispersion is economically meaningful: 22.3% of firms have ratios above 1.5, while the bottom tail includes firms for which the local first-order-condition measure points toward taxing rather than subsidizing marginal R&D.

These ratios are useful for interpreting the direction of R&D distortions, but they are not one-for-one comparable to aggregate social-return premia in reduced-form R&D-spillover estimates or to the optimal-R&D multipliers in Jones and Williams (1998) and Jones and Williams (2000). The ratio here is a firm-level, observed-state marginal-return object in the estimated equilibrium, and it nets the positive appropriability force against negative rival-profit effects from business stealing. The aggregate counterfactuals below provide the corresponding optimal-R&D comparison by solving for planner allocations, separately holding product-market allocations fixed and allowing them to adjust.

Having established the size and dispersion of the wedge, Table 7 turns to the mechanism. To interpret the table, we decompose the wedge as

$$\Delta_i(z) = \Delta_i^{\mathrm{APP}}(z) + \Delta_i^{\mathrm{RP}}(z) + \Delta_i^{\mathrm{RRC}}(z),$$

where the three terms are the appropriability (APP), rival-profit (RP), and closed-loop rival R&D resource-cost (RRC) components. This is a flow-payoff decomposition, not a decomposition by primitive network channel. The RRC component is the closed-loop rival R&D resource-cost effect, not the technology-spillover channel. Technology spillovers enter through the closed-loop state dynamics and through the solved R&D feedback rule, so they shape the discounted APP, RP, and RRC components rather than appearing as a separate component. The Lyapunov definitions of these components are in Section B.5.

Economically, APP measures the part of the household value created by firm i's marginal R&D that is not captured by aggregate producer value. It is the appropriability component of the wedge: it includes consumer surplus and other household surplus, such as labor income, generated along the induced transition path. RP measures the effect of firm i's marginal R&D on other firms' gross profits. When an innovation mainly reallocates demand or profits away from rivals, this component is negative and corresponds to business stealing. The closed-loop rival R&D resource-cost component accounts for R&D resource costs borne by other firms under their competitive feedback policies: firm i's innovation changes the future state and hence the discounted R&D resource costs borne by other firms. The signs in Table 7 follow the wedge convention: positive entries raise social marginal returns relative to private marginal returns.

The component averages show that positive appropriability forces are partly offset by negative rival-profit effects, while RRC is small. Because deciles are sorted by the social-to-private return ratio rather than by the level wedge, the mean wedge level need not be monotone across rows. The APP component is positive in every return-ratio decile and is much larger in the upper part of the distribution, increasing from 11.47 in the bottom decile to 36.58 in the ninth decile. The RP component is negative in every decile, roughly between -14.89 and -13.30, so business stealing offsets a substantial part of the positive appropriability force. The RRC component is smaller, between 0.64 and 0.87 in the baseline. Quantitatively, this channel is much smaller than the APP and RP components.

Appendix Table 11 connects these exact components to model-implied network exposure measures. The comparison shows that firms linked to more valuable spillover destinations have larger positive appropriability components, whereas firms whose R&D more strongly reallocates profits from product-market rivals have smaller appropriability components and more negative rival-profit components.

4.3 Counterfactual Allocations

The counterfactual exercises vary static product-market behavior and dynamic R&D behavior separately. Scenario labels have two letters: the first denotes the static product-market mapping and the second denotes the dynamic R&D controller; C denotes competition, M monopoly, and S the social planner. The baseline allocation is CC, with competitive product markets and competitive R&D.

The constrained counterfactuals CM and CS keep the static competitive allocation

Table 8: Scenario Comparison in 2017

Scenario	Output (CC=100)	R&D expenditure (CC=100)	Observed-state g_Y (%)	Welfare (CC=100)	Producer value share (%)
Baseline					
CC	100.00	100.00	1.48	100.00	27.09
Constrained R&D control, competitive product-market allocation					
CM	100.00	51.02	1.33	98.91	27.94
CS	100.00	230.84	1.76	100.50	25.84
Full product-market and R&D control					
MM	96.33	62.01	1.37	95.66	43.50
SS	109.29	338.69	1.83	108.97	-5.82

Notes: Columns labeled CC=100 are evaluated at the observed 2017 knowledge-capital vector and normalized to the competitive equilibrium. Total R&D expenditure is computed as $\sum_i \max\{0, x_i^u(z)\}^2$, where x_i^u is the unconstrained linear R&D rule. The observed-state growth rate is $g_Y = d \log Y_t / dt$ under the signed unconstrained drift, not a balanced-growth eigenvalue. Producer value share is $100 \times V^P/V$, net of R&D costs. It can be negative in SS because the planner maximizes household welfare rather than producer value, and discounted gross producer payoffs can be smaller than discounted R&D costs. With linear utility, the consumption-equivalent welfare change for scenario k is $100 \times (V^k/V^{CC} - 1)$, the welfare column minus 100.

fixed and change only the dynamic R&D rule. CM is therefore not a full monopoly or merger exercise; it isolates R&D-only common control, akin to a research joint venture, under competitive product-market quantities. The full monopoly allocation (MM) and full social-planner allocation (SS) let the monopolist and planner choose both production and R&D within the interior static benchmark.

This distinction is useful because product-market behavior changes the current allocation of labor and output, whereas dynamic behavior changes the incentives to accumulate knowledge capital. The log-output growth formula is derived in Section B.6, and the baseline g_Y is decomposed in Section B.7; in the competitive allocation, the 1.48% observed-state growth rate is the net of a 3.21 percentage-point contribution from technology spillovers, a 0.52 point contribution from direct R&D, and a 2.26 point drag from obsolescence. In the deterministic specification, R&D policy affects the endogenous R&D component of growth, while product-market reallocations change the output matrix that maps knowledge capital into aggregate output. Thus, comparing CC with CM or CS isolates the dynamic R&D rule, holding the competitive product-market allocation fixed. Comparing MM or SS with CC gives the effect of the full monopoly allocation or full social-planner allocation relative to the competitive equilibrium. When useful, we use CM versus MM and CS versus SS as intermediate comparisons to decompose how much of those effects comes from allowing production to be reallocated in addition to R&D.

Table 8 compiles five counterfactual allocations that retain the estimated 2017 product-market rivalry, technology-spillover networks, and knowledge-capital distribution. Output is contemporaneous output at the observed 2017 state, while total R&D expenditure and household welfare are evaluated under the scenario-specific dynamic policy. Output, total R&D expenditure, and household welfare are normalized so that the competitive equilibrium (CC) equals 100, and the producer value share reports total producer value divided by aggregate welfare. The table highlights that the constrained counterfactuals isolate the dynamic R&D margin, while the full counterfactuals add the static product-market reallocation margin.

4.3.1 Constrained R&D Control

We first examine the allocations chosen by the constrained monopolist (CM) and the constrained planner (CS), holding fixed the product-market allocation from the competitive equilibrium. With the static product-market allocation kept at its competitive level and the same initial knowledge-capital distribution, production quantities are unchanged. Consequently, contemporaneous output remains $Y = 100$ across these cases (CC, CM, CS).

Both constrained controllers internalize cross-firm effects within their objective functions, but they attach different weights to the surplus created by innovation. The monopolist (CM) evaluates innovation through aggregate producer value; it internalizes losses imposed on rival producers but ignores household value not captured by aggregate producer value. Interpreted as R&D-only common control, CM describes how an economy-wide research joint venture would choose innovation while leaving product-market quantities at their competitive levels. It cuts investment to 51.02, roughly half of the competitive equilibrium value. The constrained planner (CS) instead raises total R&D expenditure to 230.84, more than doubling the competitive equilibrium allocation, because it values innovation through household welfare, including surplus that firms do not appropriate. Because the product-market matrix $\boldsymbol{Q}$, spillover matrix $\boldsymbol{\Omega}$, obsolescence rate δ, and initial knowledge-capital vector $\boldsymbol{z}$ are held fixed across CC, CM, and CS, the difference in g_Y comes from the endogenous R&D component. Thus g_Y tracks these R&D choices: it falls to 1.33% under the monopolist, 0.15 percentage points below the competitive equilibrium, and rises to 1.76% under the constrained planner, 0.28 points above the competitive equilibrium.

Because R&D is already undersupplied in the competitive equilibrium allocation, the constrained monopolist's lower R&D expenditure pushes welfare down, leading to a 1.09% consumption-equivalent welfare loss relative to the competitive equilibrium outcome. The constrained planner, by contrast, lifts consumption-equivalent welfare 0.50% above competition. The producer value share moves in the opposite direction: prioritizing aggregate producer value raises the monopolist's share to 27.94%, whereas the constrained planner steers more surplus toward consumers and reduces the share to 25.84%. This constrained-planner result is the aggregate counterpart of the wedge measures above: many firms have social marginal R&D returns above private marginal returns, but the planner also reallocates effort across firms according to these heterogeneous gaps.

4.3.2 Full Product-Market Reallocation

We now turn to the full counterfactual allocations in which the monopolist in MM or the social planner in SS chooses both production and dynamic R&D. The main comparisons are MM and SS relative to CC, the competitive equilibrium, because these comparisons show the effect of joint control over production and innovation. The constrained allocations CM and CS then provide intermediate comparisons for decomposing how much of the full effect comes from allowing production to be reallocated in addition to R&D.

The full monopoly allocation (MM) reduces output by 3.67% relative to the competitive equilibrium allocation. The monopolist chooses static quantities to maximize current aggregate profit rather than household welfare, so it ignores household value not captured by aggregate producer value when reallocating production. By contrast, the full social-planner allocation (SS) reallocates production to maximize final-good output, so output rises by 9.29%. As noted above, these full static reallocations are unconstrained interior benchmarks; Appendix Table 19 shows that negative interior quantities arise only in MM and SS and account for less than 1 percent of the q^2 aggregate. The corresponding static $q_i \geq 0$ QPs in Appendix Table 20 change output only to 96.27 in MM and 108.82 in SS, so the main table keeps the full-reallocation rows as interior LQ benchmarks.

Changes in the product market feed directly into R&D. Relative to the competitive equilibrium allocation, the monopolist's joint choice of production and innovation in MM reduces total R&D expenditure to 62.01, roughly 38% below the competitive level. This is nevertheless higher than the 51.02 level in CM because higher markups raise the private return to new ideas. In contrast, the full social-planner allocation (SS) raises total R&D expenditure to 338.69, up from 230.84 in CS and about three and a half times the competitive equilibrium allocation. This scale is consistent with the range emphasized by Bloom et al. (2013) and with the broader optimal-R&D logic in Jones and Williams (1998, 2000). The increase from CS to SS reflects the fact that the SS static planner mapping expands output and lifts the social value of innovation.

These joint changes in production and R&D affect g_Y. Under MM, the economy grows at 1.37%, which is 0.11 percentage points below the competitive equilibrium allocation but 0.04 points above the CM outcome with fixed production. Under SS, g_Y rises to 1.83%, 0.36 points above competition and 0.08 points higher than the CS scenario. The SS reallocation accelerates knowledge accumulation, whereas the monopolist's restraint keeps growth below the competitive equilibrium allocation.

Letting both margins adjust magnifies the welfare stakes. Under MM, consumption-equivalent welfare falls 4.34% below the competitive equilibrium allocation, a much steeper loss than the 1.09% decline observed when only R&D is distorted. The extra damage comes from product-market misallocation, which erodes static efficiency on top of the innovation shortfall. In contrast, the full social-planner allocation (SS) raises consumption-equivalent welfare by 8.97% relative to competition, far exceeding the 0.50% gain secured by adjusting R&D alone. The SS static reallocation delivers an immediate boost to output and simultaneously amplifies the social return to each unit of research.

The size of this full-reallocation margin is close to the static oligopoly losses estimated with the same GHL demand structure. Pellegrino (2025) estimates that oligopoly lowers total surplus by about 11.5% for publicly traded corporations. Our SS allocation raises contemporaneous output by 9.29% and welfare by 8.97% in the 2017 patenting-firm sample. The comparison is not one-for-one because our exercise adds endogenous R&D and uses the paper's linked Compustat-DISCERN sample, but it places the large SS gain in the same quantitative range as the underlying static product-market framework.

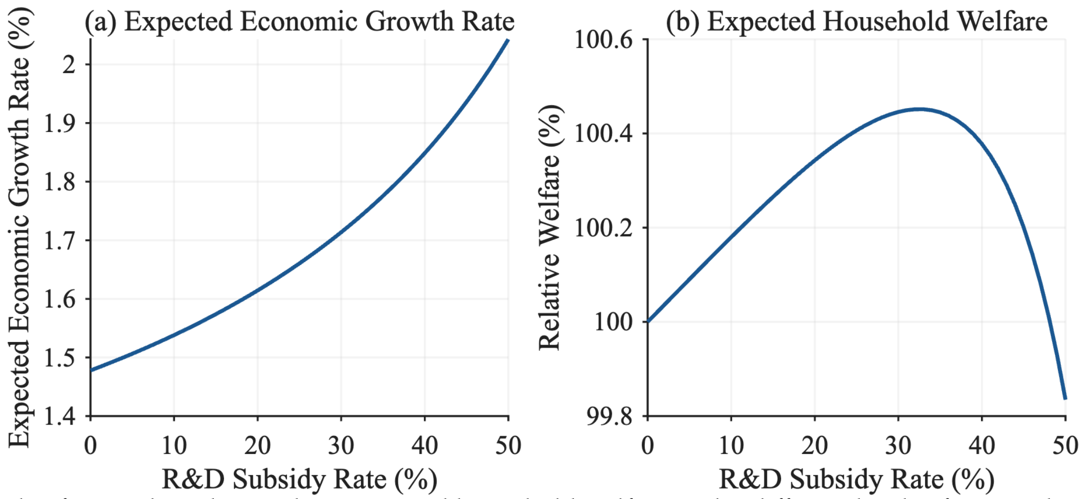
4.4 Uniform R&D Subsidy

As summarized in Table 1, a uniform R&D subsidy reduces each firm's private R&D cost from $x_{i,t}^2$ to $(1-s)x_{i,t}^2$, where $s \in [0,1)$ is the subsidy rate. The subsidy changes firms' dynamic incentives but leaves the static product-market allocation unchanged. Given a value-function coefficient matrix, the feedback rule becomes

$$\boldsymbol{x}_t = \frac{1}{1-s}\mu\widetilde{\boldsymbol{X}}(s)\boldsymbol{z}_t.$$

We treat subsidy payments as financed by lump-sum taxes, so the subsidy changes private incentives but not the aggregate resource cost of R&D; household welfare therefore continues to subtract the full resource cost $\boldsymbol{x}_t^T\boldsymbol{x}_t$. Appendix Section B.4 records the subsidy-specific Riccati and welfare equations. In the numerical exercise, we solve the competitive equilibrium over a grid of subsidy rates, $s = 0, 0.01, \dots, 0.50$, and choose the rate that maximizes household welfare. At the 2017 state, a uniform subsidy can improve welfare because aggregate equilibrium R&D falls short of the constrained planner's allocation. Welfare reaches its maximum at a subsidy of 33% on this grid (Figure 4). At that rate, g_Y rises to 1.75%, 0.27 percentage points above competition, nearly matching the constrained planner's 0.28-point increase. The subsidy also captures most of the constrained planner's

Figure 4: Observed-State Growth Rate and Household Welfare under R&D Subsidy



Notes: This figure plots observed-state g_Y and household welfare under different levels of R&D subsidy. The product-market rivalry and technology-spillover networks, as well as the distribution of knowledge capital, are estimated using data from 2017.

welfare gain on this fixed-product-market margin: it raises consumption-equivalent welfare by 0.45%, compared with 0.50% under the constrained planner. Higher subsidies eventually lower welfare as additional R&D costs outweigh growth benefits.

The wedge measures clarify the remaining allocation differences. A uniform subsidy changes the common R&D price, whereas the constrained planner's firm-level allocation reflects heterogeneous wedges. Table 9 groups firms by deciles of the baseline social-to-private return ratio and compares each decile's share of total R&D costs under competition, the solved uniform subsidy, and the constrained planner.

The uniform subsidy moves R&D shares only modestly toward the constrained-planner allocation. To summarize the remaining allocation gap, we compare each allocation's firm-level R&D cost shares with the constrained planner's shares, using one half of the sum of absolute share differences across firms. This total-variation distance is 0.0629 under competition and 0.0582 under the solved 33% uniform subsidy, a reduction of 7.5%. The reallocation vector induced by the uniform subsidy is positively correlated with the constrained planner's reallocation vector, with correlation 0.841, but the firm-level allocation remains visibly different from the constrained-planner allocation. Thus the uniform subsidy is effective for the aggregate R&D-underprovision margin in this estimated economy, while heterogeneous wedges still matter for the remaining firm-level allocation gap. This mechanism points to the value of network-based measures for assessing where uniform R&D policies leave allocation differences.

Table 9: R&D Cost Shares by Baseline Social-to-Private Return-Ratio Decile

Decile	Baseline wedge	R&D cost share (%)		
	Median SMR/PMR	Competitive	Uniform subsidy	Planner (CS)
1	0.58	0.5	0.4	0.1
2	0.94	1.2	1.1	0.5
3	1.10	1.7	1.6	0.8
4	1.19	2.4	2.3	1.4
5	1.27	3.4	3.3	2.4
6	1.35	4.4	4.3	3.4
7	1.42	7.5	7.4	6.6
8	1.48	18.4	18.4	18.0
9	1.54	41.4	41.8	44.4
10	1.66	19.1	19.2	22.4

Notes: Deciles sort firms by the baseline social-to-private return ratio, $\mathrm{SMR}_i(z)/\mathrm{PMR}_i(z)$, among firms with positive private and social marginal returns. R&D cost shares are $x_i^2/\sum_j x_j^2$ at the observed 2017 state. "Uniform" is the solved 33% competitive-subsidy equilibrium. "Planner" is the constrained-planner allocation (CS), holding the competitive product-market allocation fixed. The table is an allocation comparison, not a targeted-policy welfare estimate.

4.5 Robustness

The main scenario comparison uses $\beta = 0.02$, $\alpha = 0.12$, and $\rho = 10\%$. Appendix Sections B.8 to B.10 and Tables 12 to 14 report sensitivity exercises that recalibrate the relevant dynamic parameters and resolve the corresponding counterfactuals. The beta and product-market-parameter exercises leave the main quantitative ordering stable: the fixed-product-market planner gain remains around one half percent, the optimal uniform subsidy remains close to one third, and the full social-planner gain remains much larger. Varying the discount rate changes the level of discounted R&D gains, as expected, but preserves the ranking of the main scenarios and the finite-value checks over the reported range.

5 Conclusion

This paper develops and estimates an endogenous growth model in which many oligopolistic firms interact through empirically measured product-market rivalry and technology-spillover networks. The linear-quadratic structure reduces the dynamic R&D game to a coupled system of Riccati equations solvable at the scale of the observed firm distribution. The same apparatus can be used in other settings in which many forward-looking agents interact through measured networks.

At the observed 2017 state, competitive R&D is too low and misallocated. The median social-to-private return ratio is 1.30, with wide dispersion and a bottom tail for which the local marginal wedge points toward taxing R&D. A welfare-maximizing uniform subsidy, 33% on the grid, delivers a 0.45% consumption-equivalent welfare gain, roughly ninety percent of the constrained planner's 0.50%, because the planner's advantage comes mainly from expanding aggregate R&D rather than reallocating it across firms. R&D-only common control that maximizes producer value instead cuts R&D and lowers welfare, and full monopoly control compounds the loss. The largest gains, 8.97%, require reallocating product-market outcomes as well, a margin beyond the reach of R&D subsidies and comparable in size to static oligopoly losses estimated with the same demand structure.

The exercise has clear boundaries that define an agenda. The firm set and both networks are fixed, so entry and endogenous repositioning are absent; the spillover intensity is estimated from recovered knowledge capital rather than from quasi-experimental policy variation, so we propagate it through a sensitivity grid; and the counterfactuals are evaluated for the 2017 universe of publicly listed U.S. patenting firms. Endogenizing network formation and identifying spillover intensity from policy variation are natural next steps. The results suggest a division of labor for innovation policy: uniform instruments for the aggregate R&D margin, and network information for diagnosing where firm-level allocation differences and product-market reallocation still bind.

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Appendix

Appendix A Model Derivations

A.1 GHL Demand Reduction

The GHL demand system begins with common characteristics and idiosyncratic product characteristics. Product i is described by an m-dimensional nonnegative vector $\boldsymbol{\psi}_i$ of common characteristics, normalized so that $\boldsymbol{\psi}_i^T \boldsymbol{\psi}_i = 1$. Stacking these vectors gives

$$\boldsymbol{\Psi} \equiv \begin{bmatrix} \boldsymbol{\psi}_1 & \boldsymbol{\psi}_2 & \cdots & \boldsymbol{\psi}_n \end{bmatrix},$$

and the product-similarity matrix is

$$\boldsymbol{S} \equiv \boldsymbol{\Psi}^T \boldsymbol{\Psi}.$$

The normalization implies $S_{ii} = 1$, while for $i \neq j$, $S_{ij} = \boldsymbol{\psi}_i^T \boldsymbol{\psi}_j$ is the cosine similarity between products' common-characteristic vectors.

Let $\boldsymbol{q}_t$ be the vector of product quantities. Common and idiosyncratic characteristic quantities are

$$\boldsymbol{y}_t^C = \boldsymbol{\Psi} \boldsymbol{q}_t \tag{17}$$

and

$$\boldsymbol{y}_t^I = \boldsymbol{q}_t. \tag{18}$$

The final-good producer aggregates these characteristics according to

$$Y_t = \alpha \sum_{k=1}^{m} \left(b_{k,t}^C y_{k,t}^C - \frac{1}{2} \left(y_{k,t}^C \right)^2 \right) + (1-\alpha) \sum_{i=1}^{n} \left(b_{i,t}^I y_{i,t}^I - \frac{1}{2} \left(y_{i,t}^I \right)^2 \right), \tag{19}$$

where $\alpha \in [0, 1]$ is the weight on common characteristics. Substituting Equations (17) and (18) into Equation (19) gives

$$Y_t = \boldsymbol{q}_t^T \left[\alpha \boldsymbol{\Psi}^T \boldsymbol{b}_t^C + (1-\alpha) \boldsymbol{b}_t^I \right] - \frac{1}{2} \boldsymbol{q}_t^T \left[\alpha \boldsymbol{\Psi}^T \boldsymbol{\Psi} + (1-\alpha) \boldsymbol{I} \right] \boldsymbol{q}_t.$$

Thus the reduced-form product-quality vector and rivalry matrix are

$$\boldsymbol{b}_t \equiv \alpha \boldsymbol{\Psi}^T \boldsymbol{b}_t^C + (1-\alpha) \boldsymbol{b}_t^I, \qquad \boldsymbol{\Sigma} \equiv \alpha \boldsymbol{S} + (1-\alpha) \boldsymbol{I},$$

which yields the reduced-form aggregator in Equation (1).

A.2 Competitive Static Equilibrium

Each firm's real gross profit before subtracting R&D cost is

$$\begin{aligned}\pi_{i,t} &= \frac{p_{i,t}}{P_t} q_{i,t} - \frac{w_t}{P_t} l_{i,t} \\ &= \left(b_{i,t} - \frac{1}{a_{i,t}} \frac{w_t}{P_t} - \sum_j \sigma_{ij} q_{j,t} \right) q_{i,t},\end{aligned}$$

where the second line uses the inverse demand in Equation (6) and the production function in Equation (2). Firm i chooses $q_{i,t}$, $a_{i,t}$, and $b_{i,t}$, taking rivals' quantities, the real wage, and its own knowledge capital as given. The FOC with respect to quantity is

$$0 = b_{i,t} - \frac{1}{a_{i,t}} \frac{w_t}{P_t} - 2q_{i,t} - \sum_{j \neq i} \sigma_{ij} q_{j,t}. \tag{20}$$

The FOCs with respect to $a_{i,t}$ and $b_{i,t}$ imply

$$a_{i,t} = \sqrt{\frac{w_t}{\zeta P_t}} \tag{21}$$

$$b_{i,t} = z_{i,t} - \sqrt{\frac{\zeta w_t}{P_t}}. \tag{22}$$

Inserting Equation (21) into the labor-market-clearing condition in Equation (5) gives

$$\sqrt{\frac{\zeta w_t}{P_t}} = \frac{\zeta}{L} \sum_i q_{i,t}. \tag{23}$$

Inserting the optimality condition for technology choice Equation (21) and Equation (22) into the optimality condition for quantity choice Equation (20), and using $\sigma_{ii} = 1$ to write $2q_{i,t} + \sum_{j \neq i} \sigma_{ij} q_{j,t} = q_{i,t} + \sum_j \sigma_{ij} q_{j,t}$, we obtain

$$z_{i,t} = 2\sqrt{\zeta \frac{w_t}{P_t}} + q_{i,t} + \sum_j \sigma_{ij} q_{j,t}$$

Inserting the expression for real wage Equation (23), we obtain

$$z_{i,t} = q_{i,t} + \sum_j \left(2\frac{\zeta}{L} + \sigma_{ij}\right) q_{j,t}.$$

In vector form,

$$\boldsymbol{z}_t = \left(2\frac{\zeta}{L}\boldsymbol{J} + \boldsymbol{\Sigma} + \boldsymbol{I}\right) \boldsymbol{q}_t.$$

Therefore, quantities are linear in knowledge capital:

$$\boldsymbol{q}_t = \boldsymbol{N}\boldsymbol{z}_t$$

where

$$\boldsymbol{N} \equiv \left(2\frac{\zeta}{L}\boldsymbol{J} + \boldsymbol{\Sigma} + \boldsymbol{I}\right)^{-1}.$$

Lemma 1 (Static Cournot uniqueness)**.** *Suppose $\alpha \in [0,1]$ and $\zeta/L \geq 0$. Then $2\zeta\boldsymbol{J}/L + \boldsymbol{\Sigma} + \boldsymbol{I}$ is positive definite. Hence the interior static Cournot first-order conditions have a unique solution, given by $\boldsymbol{q}_t = \boldsymbol{N}\boldsymbol{z}_t$.*[2]

Proof. The GHL similarity matrix is $\boldsymbol{S} = \boldsymbol{\Psi}^T\boldsymbol{\Psi}$, so it is positive semidefinite. Since $\boldsymbol{\Sigma} = \alpha\boldsymbol{S} + (1-\alpha)\boldsymbol{I}$ and $\alpha \in [0,1]$, $\boldsymbol{\Sigma}$ is positive semidefinite. The all-ones matrix $\boldsymbol{J}$ is also positive semidefinite. Therefore, for any nonzero vector $\boldsymbol{v}$,

$$\boldsymbol{v}^T\left(2\frac{\zeta}{L}\boldsymbol{J} + \boldsymbol{\Sigma} + \boldsymbol{I}\right)\boldsymbol{v} = 2\frac{\zeta}{L}\boldsymbol{v}^T\boldsymbol{J}\boldsymbol{v} + \boldsymbol{v}^T\boldsymbol{\Sigma}\boldsymbol{v} + \|\boldsymbol{v}\|^2 > 0.$$

The matrix multiplying $\boldsymbol{q}_t$ in the static first-order conditions is therefore nonsingular, so the linear system has exactly one interior solution. □

Combining the quantity FOC with $\sigma_{ii} = 1$ gives

$$b_{i,t} - \frac{1}{a_{i,t}}\frac{w_t}{P_t} - \sum_j \sigma_{ij}q_{j,t} = q_{i,t},$$

so the equilibrium static gross profit is $\pi_{i,t} = q_{i,t}^2$.

[2]If nonnegativity of quantities were imposed in the static block, the same positive-definite matrix is a P-matrix, so the associated linear-complementarity problem also has a unique solution for each $\boldsymbol{z}_t$ (Cottle et al., 1992).

A.3 Output in Competitive Equilibrium

Let $\mathbf{1}$ denote an $n \times 1$ vector with all elements equal to 1. Rewrite Equation (22) in vector form:

$$\boldsymbol{b}_t = \boldsymbol{z}_t - \sqrt{\zeta \frac{w_t}{P_t}} \mathbf{1}$$

Inserting Equation (23), we obtain

$$\begin{aligned}\boldsymbol{b}_t &= \boldsymbol{z}_t - \frac{\zeta}{L} \left(\sum_i q_{i,t} \right) \mathbf{1} \\ &= \boldsymbol{z}_t - \frac{\zeta}{L} \boldsymbol{J} \boldsymbol{q}_t.\end{aligned}$$

From Equation (7), we obtain

$$\boldsymbol{b}_t = \left(\frac{\zeta}{L} \boldsymbol{J} + \boldsymbol{\Sigma} + \boldsymbol{I} \right) \boldsymbol{q}_t$$

Therefore, from Equation (1), the total output is given by

$$\begin{aligned}Y_t &= \boldsymbol{q}_t^T \left(\frac{\zeta}{L} \boldsymbol{J} + \boldsymbol{\Sigma} + \boldsymbol{I} \right) \boldsymbol{q}_t - \frac{1}{2} \boldsymbol{q}_t^T \boldsymbol{\Sigma} \boldsymbol{q}_t \\ &= \boldsymbol{q}_t^T \left(\frac{\zeta}{L} \boldsymbol{J} + \frac{1}{2} \boldsymbol{\Sigma} + \boldsymbol{I} \right) \boldsymbol{q}_t \\ &= \boldsymbol{z}_t^T \boldsymbol{N}^T \left(\frac{\zeta}{L} \boldsymbol{J} + \frac{1}{2} \boldsymbol{\Sigma} + \boldsymbol{I} \right) \boldsymbol{N} \boldsymbol{z}_t \\ &= \boldsymbol{z}_t^T \boldsymbol{Q} \boldsymbol{z}_t\end{aligned}$$

where

$$\boldsymbol{Q} \equiv \boldsymbol{N}^T \left(\frac{\zeta}{L} \boldsymbol{J} + \frac{1}{2} \boldsymbol{\Sigma} + \boldsymbol{I} \right) \boldsymbol{N}$$

A.4 Competitive Dynamic R&D Derivation

For the quadratic value-function guess in Equation (9), with $\boldsymbol{X}^i$ symmetric, the first and second derivatives are

$$\frac{\partial V^i(\boldsymbol{z})}{\partial \boldsymbol{z}} = 2 \left(\boldsymbol{X}^i \boldsymbol{z} \right)^T \tag{24}$$

$$\frac{\partial^2 V^i(\boldsymbol{z})}{\partial \boldsymbol{z} \partial \boldsymbol{z}^T} = 2 \boldsymbol{X}^i. \tag{25}$$

The derivative of Equation (8) with respect to x_i is $-2x_i + \mu \left[V_z^i(z)\right]_i = 0$. Since $\left[V_z^i(z)\right]_i = 2\left(X_i^i\right)^T z$, the first-order condition yields the linear policy in Equation (10).

The diffusion term in Equation (8) simplifies to

$$\frac{\gamma^2}{2} \operatorname{Tr}\left[\operatorname{diag}(z)\, 2X^i \operatorname{diag}(z)\right] = \gamma^2 z^T \mathcal{D}\left(X^i\right) z.$$

Substituting Equations (9), (10), (24) and (25) into Equation (8) and collecting the quadratic terms in z gives the stacked Riccati system in Equation (11).

Proof of Theorem 1. With rivals' feedback rules fixed, firm i faces a single-agent LQ control problem. Let $V^i(z) = z^T X^i z$. The Riccati equation implies that V^i satisfies firm i's HJB equation. The maximand in the HJB is strictly concave in x_i because the flow cost contains $-x_i^2$, and its first-order condition gives the feedback rule in Equation (10). For any admissible control, Itô's formula applied to $e^{-\rho t} V^i(z_t)$ on $[0, T]$ and the HJB inequality imply that the expected discounted payoff is no larger than

$$V^i(z_0) - \mathbb{E}_0\left[e^{-\rho T} V^i(z_T)\right],$$

with equality under the feedback rule above. Taking $T \to \infty$, using the admissibility condition for arbitrary controls, and using Definition 2 for the candidate feedback rule gives optimality of the candidate response. Since this argument applies to every firm, the stabilizing Riccati solution is a Markov perfect equilibrium in the stated admissible class. □

A.5 Competitive Welfare and Producer Value

After firms' equilibrium policies are fixed, household welfare is obtained from the valuation equation

$$\begin{aligned}\rho V(z) = z^T Q z - \mu^2 z^T \widetilde{X}^T \widetilde{X} z + V_z(z)\, \Phi z \\ + \frac{\gamma^2}{2} \operatorname{Tr}\left[\operatorname{diag}(z)\, V_{zz}(z) \operatorname{diag}(z)\right].\end{aligned}$$

The first two terms are household consumption, $Y_t - \sum_i x_{i,t}^2$, and the last two terms capture expected changes in continuation value. With $V(z) = z^T X z$, substituting the derivatives from Equations (24) and (25) with X^i replaced by X gives the Lyapunov equation in Equation (12).

Table 10: Growth Rate of Variables in Deterministic Economy

Growth rate	Variables
0	$l_{i,t}$
g	$a_{i,t}, b_{i,t}, z_{i,t}, q_{i,t}, x_{i,t}, p_{i,t}/P_t$
$2g$	$\pi_{i,t}, C_t, Y_t, w_t/P_t$

Notes: Growth rates are continuous-time instantaneous rates evaluated along the balanced-growth-path equilibrium in the deterministic economy with no shocks.

Aggregate producer value in the competitive equilibrium is

$$V^P(\boldsymbol{z}_0) = \mathbb{E}_0 \left[\int_0^\infty \exp(-\rho t) \left(\sum_i \pi_{i,t} - \sum_i x_{i,t}^2 \right) dt \middle| \boldsymbol{z}_0 \right].$$

Because aggregate real profit is $\sum_i \pi_{i,t} = \boldsymbol{z}_t^T \boldsymbol{N}^T \boldsymbol{N} \boldsymbol{z}_t$, the producer-value coefficient $\boldsymbol{X}^P$ solves the same Lyapunov equation as Equation (12), replacing $\boldsymbol{Q}$ with $\boldsymbol{N}^T \boldsymbol{N}$.

A.6 Deterministic Balanced Growth Path

We characterize the existence of a balanced growth path in the deterministic economy, $\gamma = 0$, using the closed-loop system $\dot{\boldsymbol{z}}_t = \boldsymbol{\Phi} \boldsymbol{z}_t$. When $\gamma > 0$, the same transition matrix governs the drift of knowledge capital. Table 10 summarizes the implied growth rates of the main variables along a deterministic balanced growth path.

We impose the following spectral condition on $\boldsymbol{\Phi}$, which is sufficient for the existence of a balanced-growth-path equilibrium.

Assumption 2. *The deterministic closed-loop technology transition matrix* $\boldsymbol{\Phi}$ *has a simple real dominant eigenvalue* $g > 0$ *with an associated strictly positive right eigenvector, and all other eigenvalues have real parts strictly below* g.

This assumption imposes the spectral property needed for balanced growth: the closed-loop technology system has a unique positive long-run growth direction. It rules out cases with multiple dominant growth directions or a non-positive balanced-growth-path knowledge-capital distribution.

Definition 3. Balanced Growth Path Equilibrium: A *balanced-growth-path equilibrium* is a linear Markov perfect equilibrium such that all firms' knowledge capital grows at the same constant rate.

Proposition 1. *Consider the deterministic economy, $\gamma = 0$. Let Assumptions 1 and 2 hold, and let $\bar{\boldsymbol{z}} \gg 0$ be the right eigenvector of $\boldsymbol{\Phi}$ associated with the dominant eigenvalue g. Then, for any scalar $c > 0$, there exists a balanced-growth-path equilibrium whose knowledge-capital path is*

$$\boldsymbol{z}_t = e^{gt} c \bar{\boldsymbol{z}}.$$

Along this path, all firms' knowledge capital grows at rate g, and the cross-sectional distribution of knowledge capital is proportional to $\bar{\boldsymbol{z}}$.

Proof. Assumption 1 gives a stabilizing solution to the stacked Riccati equations and hence linear Markov R&D policies, together with the static equilibrium mappings characterized above. In the deterministic economy, these policies imply the closed-loop system $\dot{\boldsymbol{z}}_t = \boldsymbol{\Phi} \boldsymbol{z}_t$. Since $\boldsymbol{\Phi} \bar{\boldsymbol{z}} = g \bar{\boldsymbol{z}}$, the path $\boldsymbol{z}_t = e^{gt} c \bar{\boldsymbol{z}}$ satisfies this system. Because $\bar{\boldsymbol{z}} \gg 0$ and $c > 0$, all firms' knowledge capital remains positive and grows at the common rate g. The induced static allocations and R&D policies therefore constitute a linear Markov perfect equilibrium on a balanced growth path. □

Proposition 1 implies that the equilibrium growth rate of knowledge capital is the dominant eigenvalue of $\boldsymbol{\Phi}$, and the balanced-growth-path knowledge-capital distribution is the associated eigenvector. Since $\boldsymbol{q}_t = \boldsymbol{N} \boldsymbol{z}_t$ and equilibrium R&D strategies are linear in $\boldsymbol{z}_t$, quantities and R&D grow at rate g. The static optimality conditions imply that product quality, labor productivity, and real product prices also grow at rate g, while the real wage, output, consumption, and profits grow at rate $2g$ because they are pinned down by quadratic terms in knowledge capital. In the deterministic case, Assumption 1 requires $\rho > 2g$, so discounted payoffs are finite along the balanced growth path.

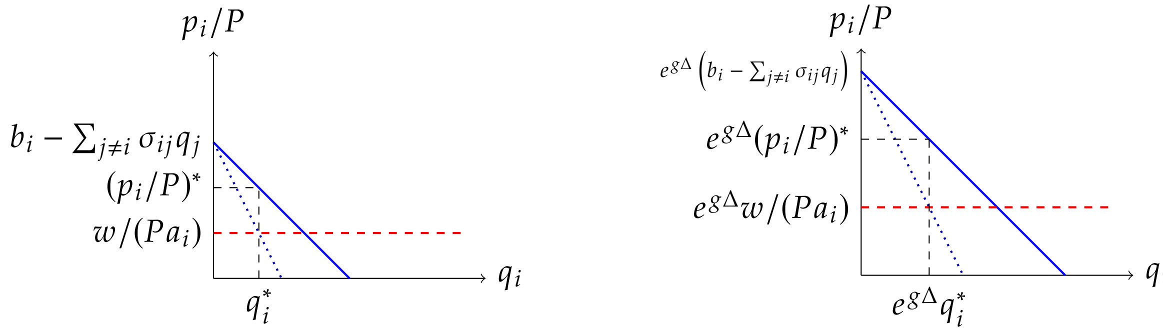
Figure 5 illustrates the mechanics of a balanced-growth-path equilibrium in the deterministic economy. In equilibrium, quality, quantity, real price, and real marginal cost grow at the same continuous-time rate g, so the partial-equilibrium diagram expands homothetically along both axes.

Appendix B Counterfactual and Wedge Derivations

B.1 Social Planner's Static Problem

We reduce the planner's static problem by substituting the knowledge-allocation constraint $b_{i,t} = z_{i,t} - \zeta a_{i,t}$ and the production technology $l_{i,t} = q_{i,t}/a_{i,t}$ into the planner's objective

Figure 5: Partial Equilibrium Diagram and Its Proportional Growth



Notes: The left panel depicts the static equilibrium for a firm's product, showing the residual demand (blue), marginal revenue (dotted line), and real marginal cost (red) curves. The right panel illustrates how the real equilibrium price, quantity, real marginal cost, and the intercept of the residual demand scale proportionally with the growth factor $\exp(g\Delta)$ over an interval of length Δ.

and labor-market constraint. Define the Lagrangian as

$$\mathcal{L} = \boldsymbol{q}_t^T (\boldsymbol{z}_t - \zeta \boldsymbol{a}_t) - \frac{1}{2}\boldsymbol{q}_t^T \boldsymbol{\Sigma} \boldsymbol{q}_t - \lambda_t \left\{ \sum_i \frac{q_{i,t}}{a_{i,t}} - L \right\}$$

We focus on the interior solution in which the labor constraint binds. The FOC for $a_{i,t}$ implies $\lambda_t = \zeta a_{i,t}^2 > 0$, so the square root below is well defined. The FOC w.r.t. $a_{i,t}$ gives

$$a_{i,t} = \sqrt{\frac{\lambda_t}{\zeta}} \tag{26}$$

and therefore,

$$b_{i,t} = z_{i,t} - \sqrt{\zeta \lambda_t} \tag{27}$$

Inserting Equation (26) into Equation (13), we obtain

$$\sqrt{\zeta \lambda_t} = \frac{\zeta}{L} \sum_i q_{i,t}$$

Note that

$$\sqrt{\zeta \lambda_t}\boldsymbol{1} = \frac{\zeta}{L}\left(\sum_i q_{i,t}\right)\boldsymbol{1} = \frac{\zeta}{L}\boldsymbol{J}\boldsymbol{q}_t. \tag{28}$$

The FOC w.r.t. $\boldsymbol{q}_t$ gives

$$\boldsymbol{z}_t - \zeta \boldsymbol{a}_t - \boldsymbol{\Sigma} \boldsymbol{q}_t = \lambda_t \begin{bmatrix} 1/a_{1,t} \\ 1/a_{2,t} \\ \vdots \\ 1/a_{n,t} \end{bmatrix}$$

Substituting Equation (26), we obtain

$$\boldsymbol{z}_t = 2\sqrt{\zeta \lambda_t} \mathbf{1} + \boldsymbol{\Sigma} \boldsymbol{q}_t$$

Using Equation (28), we obtain

$$\boldsymbol{z}_t = \left(2 \frac{\zeta}{L} \boldsymbol{J} + \boldsymbol{\Sigma} \right) \boldsymbol{q}_t$$

Therefore,

$$\boldsymbol{q}_t = \boldsymbol{N}^* \boldsymbol{z}_t \tag{29}$$

where

$$\boldsymbol{N}^* = \left(2 \frac{\zeta}{L} \boldsymbol{J} + \boldsymbol{\Sigma} \right)^{-1}$$

Because $\boldsymbol{J}$ and $\boldsymbol{\Sigma}$ are symmetric, $\boldsymbol{N}^*$ is symmetric as well. To obtain the quadratic form of output, substitute Equation (27) into Equation (1):

$$Y_t = \boldsymbol{q}_t^T \left(\boldsymbol{z}_t - \sqrt{\zeta \lambda_t} \mathbf{1} \right) - \frac{1}{2} \boldsymbol{q}_t^T \boldsymbol{\Sigma} \boldsymbol{q}_t$$

Using Equation (28) together with Equation (29),

$$\begin{aligned} Y_t &= \boldsymbol{q}_t^T \left(\left(2 \frac{\zeta}{L} \boldsymbol{J} + \boldsymbol{\Sigma} \right) \boldsymbol{q}_t - \frac{\zeta}{L} \boldsymbol{J} \boldsymbol{q}_t \right) - \frac{1}{2} \boldsymbol{q}_t^T \boldsymbol{\Sigma} \boldsymbol{q}_t \\ &= \boldsymbol{q}_t^T \left(\frac{\zeta}{L} \boldsymbol{J} + \frac{1}{2} \boldsymbol{\Sigma} \right) \boldsymbol{q}_t \\ &= \frac{1}{2} \boldsymbol{q}_t^T (\boldsymbol{N}^*)^{-1} \boldsymbol{q}_t \\ &= \frac{1}{2} \boldsymbol{z}_t^T \boldsymbol{N}^* \boldsymbol{z}_t \end{aligned}$$

B.2 Planner Dynamic R&D Derivation

The HJB equation for the social planner's dynamic problem is

$$\rho V^{SS}(\boldsymbol{z}) = \max_{\boldsymbol{x}} \left\{ \frac{1}{2}\boldsymbol{z}^T \boldsymbol{N}^* \boldsymbol{z} - \boldsymbol{x}^T \boldsymbol{x} + V_z^{SS}(\boldsymbol{z}) \left[(\boldsymbol{\Omega} - \delta \boldsymbol{I}) \boldsymbol{z} + \mu \boldsymbol{x} \right] \right.$$
$$\left. + \frac{\gamma^2}{2} \mathrm{Tr} \left[\mathrm{diag}(\boldsymbol{z}) V_{zz}^{SS}(\boldsymbol{z}) \mathrm{diag}(\boldsymbol{z}) \right] \right\}. \tag{30}$$

Using the quadratic guess

$$V^{SS}(\boldsymbol{z}) = \boldsymbol{z}^T \boldsymbol{X}^{SS} \boldsymbol{z}, \tag{31}$$

the first-order condition with respect to $\boldsymbol{x}$ is $-2\boldsymbol{x} + \mu \left(V_z^{SS}(\boldsymbol{z}) \right)^T = 0$, which gives the linear R&D rule in Equation (14). The induced transition is $\boldsymbol{\Phi}^{SS} = \boldsymbol{\Omega} + \mu^2 \boldsymbol{X}^{SS} - \delta \boldsymbol{I}$. We take the value-function coefficient matrix to be symmetric, since only its symmetric part affects $\boldsymbol{z}^T \boldsymbol{X}^{SS} \boldsymbol{z}$. Thus the drift term is represented by $\boldsymbol{X}^{SS} \boldsymbol{\Phi}^{SS} + (\boldsymbol{\Phi}^{SS})^T \boldsymbol{X}^{SS}$ in quadratic form. This path-flow convention matters because the estimated spillover matrix $\boldsymbol{\Omega}$ is generally asymmetric. Substituting the quadratic value function, the linear R&D rule, and this transition into Equation (30) gives the planner Riccati equation in Equation (15).

B.3 Monopoly Allocation and Welfare

This subsection derives the full monopoly allocation (MM) used in the counterfactual analysis. We keep the superscript M for static monopoly matrices, such as $\boldsymbol{N}^M$, $\boldsymbol{P}^M$, and $\boldsymbol{Q}^M$, and use the superscript MM for the full scenario in which the monopolist controls both production and R&D. Since one decision maker controls the full R&D vector, the monopoly R&D policy is obtained from a single Riccati equation; household welfare under that policy is then computed separately from a Lyapunov equation.

Static Mapping

The monopolist chooses $\boldsymbol{a}_t$, $\boldsymbol{b}_t$, and $\boldsymbol{q}_t$ to maximize aggregate real profit. Differentiating Π_t with respect to $q_{i,t}$, and using the symmetry of $\boldsymbol{\Sigma}$ so that $\partial(\boldsymbol{q}_t^T \boldsymbol{\Sigma} \boldsymbol{q}_t)/\partial q_{i,t} = 2\sum_j \sigma_{ij} q_{j,t}$, gives

$$0 = b_{i,t} - \frac{1}{a_{i,t}} \frac{w_t}{P_t} - 2\sum_j \sigma_{ij} q_{j,t}. \tag{32}$$

The real profit of each product is therefore

$$\begin{aligned}\pi_{i,t} &= \left(b_{i,t} - \frac{1}{a_{i,t}}\frac{w_t}{P_t}\right) q_{i,t} - \sum_j \sigma_{ij} q_{i,t} q_{j,t} \\ &= 2\sum_j \sigma_{ij} q_{j,t} q_{i,t} - \sum_j \sigma_{ij} q_{i,t} q_{j,t} \\ &= \sum_j \sigma_{ij} q_{i,t} q_{j,t}. \end{aligned} \tag{33}$$

The FOC with respect to $a_{i,t}$ gives

$$a_{i,t} = \sqrt{\frac{1}{\zeta}\frac{w_t}{P_t}}, \tag{34}$$

and therefore,

$$b_{i,t} = z_{i,t} - \sqrt{\zeta \frac{w_t}{P_t}}. \tag{35}$$

Inserting Equation (34) into the labor-market-clearing condition Equation (5) gives

$$\sqrt{\zeta \frac{w_t}{P_t}} = \frac{\zeta}{L} \sum_i q_{i,t}. \tag{36}$$

Combining Equations (32), (35) and (36) yields

$$\boldsymbol{z}_t = 2\left(\frac{\zeta}{L}\boldsymbol{J} + \boldsymbol{\Sigma}\right)\boldsymbol{q}_t, \tag{37}$$

so monopoly quantities are linear in knowledge capital:

$$\boldsymbol{q}_t^M = \boldsymbol{N}^M \boldsymbol{z}_t, \qquad \boldsymbol{N}^M \equiv \frac{1}{2}\left(\frac{\zeta}{L}\boldsymbol{J} + \boldsymbol{\Sigma}\right)^{-1}. \tag{38}$$

From Equation (33), aggregate real profit is

$$\Pi_t^M = \boldsymbol{q}_t^T \boldsymbol{\Sigma} \boldsymbol{q}_t = \boldsymbol{z}_t^T \boldsymbol{P}^M \boldsymbol{z}_t, \qquad \boldsymbol{P}^M \equiv (\boldsymbol{N}^M)^T \boldsymbol{\Sigma} \boldsymbol{N}^M.$$

Output and Producer Surplus

As in the competitive equilibrium,

$$\boldsymbol{b}_t = \boldsymbol{z}_t - \frac{\zeta}{L}\boldsymbol{J}\boldsymbol{q}_t.$$

Using Equation (37), this becomes

$$\boldsymbol{b}_t = \left(\frac{\zeta}{L}\boldsymbol{J} + 2\boldsymbol{\Sigma}\right)\boldsymbol{q}_t.$$

Therefore, from Equation (1), monopoly output is

$$\begin{aligned} Y_t^M &= \boldsymbol{q}_t^T \left(\frac{\zeta}{L}\boldsymbol{J} + 2\boldsymbol{\Sigma}\right)\boldsymbol{q}_t - \frac{1}{2}\boldsymbol{q}_t^T\boldsymbol{\Sigma}\boldsymbol{q}_t \\ &= \frac{1}{2}\boldsymbol{q}_t^T\left(2\frac{\zeta}{L}\boldsymbol{J} + 3\boldsymbol{\Sigma}\right)\boldsymbol{q}_t = \boldsymbol{z}_t^T\boldsymbol{Q}^M\boldsymbol{z}_t, \end{aligned}$$

where

$$\boldsymbol{Q}^M \equiv \frac{1}{2}(\boldsymbol{N}^M)^T\left(2\frac{\zeta}{L}\boldsymbol{J} + 3\boldsymbol{\Sigma}\right)\boldsymbol{N}^M.$$

Dynamic R&D

The monopolist chooses R&D to maximize aggregate producer value:

$$V^{P,MM}(\boldsymbol{z}_0) \equiv \max_{\{\boldsymbol{x}_t\}_{t\geq 0}} \mathbb{E}_0\left[\int_0^\infty \exp(-\rho t)\left\{\boldsymbol{z}_t^T\boldsymbol{P}^M\boldsymbol{z}_t - \boldsymbol{x}_t^T\boldsymbol{x}_t\right\}dt\right],$$

subject to the knowledge accumulation law in Equation (4). We obtain the monopoly R&D rule by solving Equation (15) with $\frac{1}{2}\boldsymbol{N}^*$ replaced by $\boldsymbol{P}^M$. Let $\boldsymbol{X}^{P,MM}$ denote the symmetric coefficient matrix in the monopolist's producer-value function. The R&D allocation is

$$\boldsymbol{x}_t = \mu\boldsymbol{X}^{P,MM}\boldsymbol{z}_t,$$

and the induced transition is

$$\boldsymbol{\Phi}^{MM} \equiv \boldsymbol{\Omega} - \delta\boldsymbol{I} + \mu^2\boldsymbol{X}^{P,MM}.$$

Household Welfare

In contrast to the social planner, the monopolist maximizes aggregate producer value rather than household utility. Hence household welfare under the monopoly policy is computed separately:

$$V^{MM}\left(\boldsymbol{z}_0\right)=\mathbb{E}_0\left[\int_0^{\infty}\exp\left(-\rho t\right)\left(\boldsymbol{z}_t^T\boldsymbol{Q}^M\boldsymbol{z}_t-\boldsymbol{x}_t^T\boldsymbol{x}_t\right)dt\middle|\boldsymbol{z}_0\right],$$

subject to $d\boldsymbol{z}_t=\boldsymbol{\Phi}^{MM}\boldsymbol{z}_t dt+\gamma\operatorname{diag}\left(\boldsymbol{z}_t\right)d\boldsymbol{W}_t$. Guessing $V^{MM}(\boldsymbol{z})=\boldsymbol{z}^T\boldsymbol{X}^{MM}\boldsymbol{z}$ gives the Lyapunov equation

$$\begin{aligned}0=\boldsymbol{Q}^M-\mu^2\left(\boldsymbol{X}^{P,MM}\right)^T\boldsymbol{X}^{P,MM}+\boldsymbol{X}^{MM}\left(\boldsymbol{\Phi}^{MM}-\frac{\rho}{2}\boldsymbol{I}\right)\\+\left(\boldsymbol{\Phi}^{MM}-\frac{\rho}{2}\boldsymbol{I}\right)^T\boldsymbol{X}^{MM}+\gamma^2\mathcal{D}\left(\boldsymbol{X}^{MM}\right).\end{aligned}\tag{39}$$

Therefore, $\boldsymbol{X}^{MM}$ is obtained as the solution of Equation (39), and welfare in the monopoly problem is $V^{MM}(\boldsymbol{z}_0)=\boldsymbol{z}_0^T\boldsymbol{X}^{MM}\boldsymbol{z}_0$.

B.4 Uniform R&D Subsidy Equations

This appendix records the equations used in the uniform R&D subsidy exercise. Let $s\in[0,1)$ denote a subsidy rate that reduces the private cost of R&D effort from $x_{i,t}^2$ to $(1-s)x_{i,t}^2$. Holding the static product-market allocation fixed, firm i solves

$$V^i\left(\boldsymbol{z}_0\right)=\max_{\{x_{i,t}\}_{t\geq 0}}\mathbb{E}_0\left[\int_0^{\infty}e^{-\rho t}\left\{\boldsymbol{z}_t^T\boldsymbol{Q}_i\boldsymbol{z}_t-(1-s)x_{i,t}^2\right\}dt\right],$$

taking other firms' R&D rules as given. With the quadratic guess $V^i(\boldsymbol{z})=\boldsymbol{z}^T\boldsymbol{X}^i\boldsymbol{z}$, the first-order condition is

$$x_i=\frac{\mu}{1-s}\left(\boldsymbol{X}_i^i\right)^T\boldsymbol{z}.$$

Stacking firms' policies into the subsidy-specific policy matrix $\widetilde{\boldsymbol{X}}(s)$ gives

$$\boldsymbol{x}_t=\frac{\mu}{1-s}\widetilde{\boldsymbol{X}}(s)\boldsymbol{z}_t,\qquad\boldsymbol{\Phi}(s)=\boldsymbol{\Omega}-\delta\boldsymbol{I}+\frac{\mu^2}{1-s}\widetilde{\boldsymbol{X}}(s).$$

The competitive Riccati equations under the subsidy become

$$0 = \boldsymbol{Q}_i - \frac{\mu^2}{1-s}\boldsymbol{X}_i^i \left(\boldsymbol{X}_i^i\right)^T + \left(\boldsymbol{\Phi}(s) - \frac{\rho}{2}\boldsymbol{I}\right)^T \boldsymbol{X}^i + \boldsymbol{X}^i \left(\boldsymbol{\Phi}(s) - \frac{\rho}{2}\boldsymbol{I}\right) + \gamma^2 \mathcal{D}\left(\boldsymbol{X}^i\right).$$

The subsidy changes firms' private cost but not the resource cost borne by the household. Hence household welfare subtracts the full cost

$$\boldsymbol{x}_t^T \boldsymbol{x}_t = \frac{\mu^2}{(1-s)^2} \boldsymbol{z}_t^T \widetilde{\boldsymbol{X}}(s)^T \widetilde{\boldsymbol{X}}(s) \boldsymbol{z}_t.$$

Given the equilibrium policy, household welfare is obtained from the Lyapunov equation

$$0 = \boldsymbol{Q} - \frac{\mu^2}{(1-s)^2}\widetilde{\boldsymbol{X}}(s)^T \widetilde{\boldsymbol{X}}(s) + \boldsymbol{X}(s)\left(\boldsymbol{\Phi}(s) - \frac{\rho}{2}\boldsymbol{I}\right) + \left(\boldsymbol{\Phi}(s) - \frac{\rho}{2}\boldsymbol{I}\right)^T \boldsymbol{X}(s) + \gamma^2 \mathcal{D}\left(\boldsymbol{X}(s)\right).$$

The numerical exercise solves this system for $s = 0, 0.01, \ldots, 0.50$ and selects the value of s that maximizes $\boldsymbol{z}_0^T \boldsymbol{X}(s) \boldsymbol{z}_0$ at the 2017 knowledge-capital vector.

The main text reports the allocation comparison between R&D cost shares under competition, the solved 33% uniform subsidy, and the constrained planner.

B.5 Social-Private R&D Wedge Decomposition

This appendix gives the component definitions used in Table 7. Let $\boldsymbol{e}_i$ denote the ith standard basis vector. Let $\boldsymbol{P}$ denote the aggregate competitive gross-profit matrix,

$$\boldsymbol{P} \equiv \sum_{j=1}^{n} \boldsymbol{Q}_j = \boldsymbol{N}^T \boldsymbol{N}.$$

Thus aggregate competitive gross profit is $\sum_j \pi_j(\boldsymbol{z}) = \boldsymbol{z}^T \boldsymbol{P} \boldsymbol{z}$. The social-private R&D wedge can be written as

$$\Delta_i(\boldsymbol{z}) = 2\mu\, \boldsymbol{e}_i^T \left(\boldsymbol{X} - \boldsymbol{X}^i\right) \boldsymbol{z}. \tag{40}$$

Define $\boldsymbol{M}_i \equiv \boldsymbol{X} - \boldsymbol{X}^i$ and

$$\boldsymbol{C}_{-i} \equiv \widetilde{\boldsymbol{X}}^T \left(\boldsymbol{I} - \boldsymbol{e}_i \boldsymbol{e}_i^T\right) \widetilde{\boldsymbol{X}}.$$

Since the competitive feedback rule is $\boldsymbol{x}^{C}(\boldsymbol{z}) = \mu \widetilde{\boldsymbol{X}} \boldsymbol{z}$, the quadratic form $\mu^2 \boldsymbol{C}_{-i}$ gives the R&D resource costs of all firms other than i under the competitive policy. Subtracting firm i's Riccati equation from the household welfare Lyapunov equation gives

$$\mathbf{0} = \boldsymbol{F}_i + \boldsymbol{M}_i \left(\boldsymbol{\Phi} - \frac{\rho}{2} \boldsymbol{I} \right) + \left(\boldsymbol{\Phi} - \frac{\rho}{2} \boldsymbol{I} \right)^T \boldsymbol{M}_i + \gamma^2 \mathcal{D}(\boldsymbol{M}_i), \tag{41}$$

where the flow wedge is

$$\begin{aligned} \boldsymbol{F}_i &\equiv \boldsymbol{Q} - \boldsymbol{Q}_i - \mu^2 \boldsymbol{C}_{-i} \\ &= \underbrace{\boldsymbol{Q} - \boldsymbol{P}}_{\text{appropriability}} + \underbrace{\boldsymbol{P} - \boldsymbol{Q}_i}_{\text{rival profit}} \\ &\quad - \underbrace{\mu^2 \boldsymbol{C}_{-i}}_{\text{rival R\&D cost}} . \end{aligned} \tag{42}$$

This decomposition separates the flow-payoff consequences of a marginal increase in firm i's knowledge drift under the competitive closed-loop policy. The term $\boldsymbol{Q} - \boldsymbol{P}$ is the household output value that is not captured by aggregate producer gross profit, so it is the appropriability component; as noted in Section 2.2.5, this is the static output-profit gap $\boldsymbol{Q} - \boldsymbol{N}^T \boldsymbol{N}$. The term $\boldsymbol{P} - \boldsymbol{Q}_i$ is the gross-profit flow accruing to firms other than i; the reported RP component is its marginal value with respect to z_i, which is negative when higher z_i lowers rivals' quantities. Finally, $-\mu^2 \boldsymbol{C}_{-i}$ subtracts the R&D resource costs borne by other firms under their competitive feedback rules. The sign of the resulting RRC marginal component depends on the closed-loop feedback response; the positive average entries in Table 7 indicate that the marginal state change lowers discounted rival R&D resource costs in the baseline decile groups. These components decompose flow payoffs rather than primitive network channels: technology spillovers enter through the closed-loop state dynamics and the solved R&D feedback rule, so they shape the discounted APP, RP, and RRC components rather than appearing as a separate component.

For each component $m \in \{\text{APP}, \text{RP}, \text{RRC}\}$, solve Equation (41) with the corresponding flow term

$$\boldsymbol{F}^{\text{APP}} = \boldsymbol{Q} - \boldsymbol{P}, \qquad \boldsymbol{F}_i^{\text{RP}} = \boldsymbol{P} - \boldsymbol{Q}_i, \qquad \boldsymbol{F}_i^{\text{RRC}} = -\mu^2 \boldsymbol{C}_{-i}.$$

Denoting the resulting solutions by $\boldsymbol{M}^{\text{APP}}$, $\boldsymbol{M}_i^{\text{RP}}$, and $\boldsymbol{M}_i^{\text{RRC}}$, the component wedges are

$$\begin{aligned} \Delta_i^{\text{APP}}(\boldsymbol{z}) &= 2\mu \, \boldsymbol{e}_i^T \boldsymbol{M}^{\text{APP}} \boldsymbol{z}, \\ \Delta_i^{\text{RP}}(\boldsymbol{z}) &= 2\mu \, \boldsymbol{e}_i^T \boldsymbol{M}_i^{\text{RP}} \boldsymbol{z}, \end{aligned} \tag{43}$$

$$\Delta_i^{\mathrm{RRC}}(\boldsymbol{z}) = 2\mu\, \boldsymbol{e}_i^T \boldsymbol{M}_i^{\mathrm{RRC}} \boldsymbol{z},$$

Because Equation (41) is linear in the pair $(\boldsymbol{M}, \boldsymbol{F})$, the component solutions sum to $\boldsymbol{M}_i$, so the component wedges add up exactly to the total wedge:

$$\Delta_i(\boldsymbol{z}) = \Delta_i^{\mathrm{APP}}(\boldsymbol{z}) + \Delta_i^{\mathrm{RP}}(\boldsymbol{z}) + \Delta_i^{\mathrm{RRC}}(\boldsymbol{z}). \tag{44}$$

Component-Level Network Analysis

For the component-level network analysis, define two firm-level flow-proxy exposures, evaluated at the observed 2017 knowledge-capital vector before the regression standardization:

$$\text{Technology-spillover exposure}_i(\boldsymbol{z}) \equiv 2 \sum_{j \neq i} \Omega_{ji}\, \boldsymbol{e}_j^T \boldsymbol{X} \boldsymbol{z}, \tag{45}$$

$$\text{Business-stealing exposure}_i(\boldsymbol{z}) \equiv -\sum_{j \neq i} \frac{\partial \pi_j(\boldsymbol{z})}{\partial z_i} = -\sum_{j \neq i} 2 q_j N_{ji}. \tag{46}$$

Because rows of $\boldsymbol{\Omega}$ are recipients, $[\boldsymbol{\Omega}]_{ji}$ is the effective spillover exposure of recipient j to source firm i. The value weight $2\boldsymbol{e}_j^T \boldsymbol{X}\boldsymbol{z}$ is the household-welfare marginal value of recipient j's knowledge capital, so the measure emphasizes spillovers to recipients whose knowledge is more valuable in the model. Since $\boldsymbol{q} = \boldsymbol{N}\boldsymbol{z}$ and $\pi_j = q_j^2$, business-stealing exposure measures the extent to which an increase in firm i's knowledge capital reduces rivals' gross profits.

Table 11 reports a component-level network analysis for the same observed-state competitive-policy wedge using the exposures in Equations (45) and (46). The regressions relate the exact Lyapunov components to technology-spillover exposure and business-stealing exposure, controlling for firm knowledge capital.

Two patterns are useful for interpreting the decomposition. First, technology-spillover exposure is positively associated with both the appropriability component and the rival-profit component. Thus, firms whose R&D reaches high-value technology recipients tend to generate larger appropriability gains, and their spillovers also partly offset the rival-profit losses generated through product-market competition. Second, business-stealing exposure is negatively associated with both the appropriability and rival-profit components. The association is strongest for RP, consistent with the direct competitive channel: firms whose R&D more strongly reallocates sales from product-market rivals have more negative rival-profit effects. The negative APP coefficient shows that business-stealing-intensive

Table 11: Network Correlates of Social-Private R&D Wedge Components

	Total wedge	APP	RP	RRC
Technology-spillover exposure	0.305	0.235	0.684	-0.964
Business-stealing exposure	-0.232	-0.125	-0.971	0.620
log z	0.814	0.822	-0.041	0.060
R^2	0.848	0.844	0.905	0.824
Observations	739	739	739	739

Notes: The table reports standardized coefficients from flow-proxy regressions for the 739-firm positive marginal-return sample at the observed 2017 state. Dependent variables are the exact deterministic Lyapunov wedge components defined in Section B.5; both dependent variables and predictors are standardized. Each coefficient is therefore the standard-deviation change in the wedge component associated with a one-standard-deviation increase in the exposure, conditional on the other regressors. Technology-spillover exposure and business-stealing exposure are defined in Equations (45) and (46). APP is the appropriability component, RP is the rival-profit component, and RRC is the closed-loop rival R&D resource-cost component. These regressions are descriptive model-internal correlations rather than causal estimates. The RRC component remains a secondary mechanism because its level contribution is small in Table 7.

R&D is also less likely to generate broad aggregate propagation gains.

B.6 Expected Log Output Growth

This subsection derives the log-output growth formula used in the baseline observed-state decomposition below. Apply Itô's lemma to the closed-loop knowledge law

$$d\boldsymbol{z}_t = \boldsymbol{\Phi}\boldsymbol{z}_t dt + \gamma \,\mathrm{diag}\left(\boldsymbol{z}_t\right) d\boldsymbol{W}_t$$

and to output $Y_t = f(\boldsymbol{z}_t) = \boldsymbol{z}_t^T \boldsymbol{Q} \boldsymbol{z}_t$. The matrix $\boldsymbol{Q}$ is symmetric by construction, and the log-growth formula below is evaluated on states with $Y_t > 0$. The required derivatives are

$$\begin{aligned} f_z\left(\boldsymbol{z}_t\right) &= 2\boldsymbol{Q}\boldsymbol{z}_t, \\ f_{zz}\left(\boldsymbol{z}_t\right) &= 2\boldsymbol{Q}. \end{aligned}$$

Therefore,

$$\begin{aligned} dY_t &= \left\{2\boldsymbol{z}_t^T\boldsymbol{Q}\boldsymbol{\Phi}\boldsymbol{z}_t + \frac{1}{2}\mathrm{Tr}\left[2\gamma^2\,\mathrm{diag}\left(\boldsymbol{z}_t\right)\boldsymbol{Q}\,\mathrm{diag}\left(\boldsymbol{z}_t\right)\right]\right\}dt + 2\gamma\boldsymbol{z}_t^T\boldsymbol{Q}\,\mathrm{diag}\left(\boldsymbol{z}_t\right)d\boldsymbol{W}_t \\ &= \left\{\boldsymbol{z}_t^T\left(\boldsymbol{Q}\boldsymbol{\Phi} + \boldsymbol{\Phi}^T\boldsymbol{Q}\right)\boldsymbol{z}_t + \gamma^2\sum_i z_{i,t}^2\boldsymbol{Q}_{ii}\right\}dt + 2\gamma\boldsymbol{z}_t^T\boldsymbol{Q}\,\mathrm{diag}\left(\boldsymbol{z}_t\right)d\boldsymbol{W}_t \end{aligned}$$

Next, applying Itô's lemma to $g(Y_t) = \log Y_t$ and using the diffusion term in dY_t gives

$$d \log Y_t = \left[\frac{\boldsymbol{z}_t^T \left(\boldsymbol{Q\Phi} + \boldsymbol{\Phi}^T \boldsymbol{Q}\right) \boldsymbol{z}_t}{\boldsymbol{z}_t^T \boldsymbol{Q} \boldsymbol{z}_t} + \gamma^2 \left\{ \frac{\sum_i z_{i,t}^2 \boldsymbol{Q}_{ii}}{\boldsymbol{z}_t^T \boldsymbol{Q} \boldsymbol{z}_t} - 2 \frac{\boldsymbol{z}_t^T \boldsymbol{Q} \operatorname{diag}\left(\boldsymbol{z}_t^2\right) \boldsymbol{Q} \boldsymbol{z}_t}{\left(\boldsymbol{z}_t^T \boldsymbol{Q} \boldsymbol{z}_t\right)^2} \right\} \right] dt$$
$$+ \frac{2\gamma}{\boldsymbol{z}_t^T \boldsymbol{Q} \boldsymbol{z}_t} \boldsymbol{z}_t^T \boldsymbol{Q} \operatorname{diag}(\boldsymbol{z}_t)\, d\boldsymbol{W}_t.$$

Because the technology transition matrix can be written as

$$\boldsymbol{\Phi} = \boldsymbol{\Omega} - \delta \boldsymbol{I} + \mu^2 \widetilde{\boldsymbol{X}},$$

the first drift term in log output can be decomposed as

$$\boldsymbol{Q\Phi} + \boldsymbol{\Phi}^T \boldsymbol{Q} = \left(\boldsymbol{Q\Omega} + \boldsymbol{\Omega}^T \boldsymbol{Q}\right) + \mu^2 \left(\boldsymbol{Q}\widetilde{\boldsymbol{X}} + \widetilde{\boldsymbol{X}}^T \boldsymbol{Q}\right) - 2\delta \boldsymbol{Q}.$$

Thus, the contributions of technology spillovers, R&D, obsolescence, and the Itô term to observed-state g_Y are

$$\frac{d \log Y_t|_{\text{spillover}}}{dt} = \frac{\boldsymbol{z}_t^T \left(\boldsymbol{Q\Omega} + \boldsymbol{\Omega}^T \boldsymbol{Q}\right) \boldsymbol{z}_t}{\boldsymbol{z}_t^T \boldsymbol{Q} \boldsymbol{z}_t} \tag{47}$$

$$\frac{d \log Y_t|_{\text{R\&D}}}{dt} = \mu^2 \frac{\boldsymbol{z}_t^T \left(\boldsymbol{Q}\widetilde{\boldsymbol{X}} + \widetilde{\boldsymbol{X}}^T \boldsymbol{Q}\right) \boldsymbol{z}_t}{\boldsymbol{z}_t^T \boldsymbol{Q} \boldsymbol{z}_t} \tag{48}$$

$$\frac{d \log Y_t|_{\text{Obsolescence}}}{dt} = -2\delta \tag{49}$$

$$\frac{d \log Y_t|_{\text{Itô}}}{dt} = \gamma^2 \left\{ \frac{\sum_i z_{i,t}^2 \boldsymbol{Q}_{ii}}{\boldsymbol{z}_t^T \boldsymbol{Q} \boldsymbol{z}_t} - 2 \frac{\boldsymbol{z}_t^T \boldsymbol{Q} \operatorname{diag}\left(\boldsymbol{z}_t^2\right) \boldsymbol{Q} \boldsymbol{z}_t}{\left(\boldsymbol{z}_t^T \boldsymbol{Q} \boldsymbol{z}_t\right)^2} \right\}.$$

B.7 Baseline Observed-State Growth Decomposition

To interpret the g_Y column in the counterfactual table, we decompose the baseline observed-state growth rate into technology spillovers, obsolescence, R&D, and the contribution of shocks. The components are computed from the primitive matrices $\boldsymbol{Q}$, $\boldsymbol{\Omega}$, $\widetilde{\boldsymbol{X}}$, and the observed 2017 knowledge-capital vector so that they add up to g_Y under the baseline policy.

The baseline observed-state growth rate is 1.48%, close to the growth target used in the calibration. It is the net of a 3.21 percentage-point contribution from technology spillovers, a 0.52 point contribution from direct R&D, and a 2.26 point drag from knowledge obsolescence. In the baseline deterministic specification, the Itô term is zero because

Table 12: Spillover-Intensity Sensitivity

β	μ	δ	CC g_Y components, pp			Welfare		Uniform subsidy	
			Spill.	R&D	Dep.	CS welfare	SS welfare	Subsidy rate (%)	Subsidy welfare
0.000	0.0506	0.0000	0.00	0.55	-0.00	100.45	112.14	27	100.35
0.010	0.0436	0.0017	1.61	0.41	-0.35	100.32	109.68	29	100.28
0.020	0.0525	0.0113	3.21	0.52	-2.26	100.50	108.97	33	100.45
0.030	0.0570	0.0194	4.82	0.56	-3.87	100.76	108.36	37	100.68

Notes: For each β, μ and δ are recalibrated in the 2017 competitive equilibrium to match aggregate R&D intensity of 2.6% and observed-state output growth g_Y of 1.5%. Spill., R&D, and Dep. decompose competitive-equilibrium g_Y in percentage points. CS, SS, and uniform-subsidy welfare are normalized to the competitive equilibrium at the same β. The uniform-subsidy rate is the welfare-maximizing R&D subsidy on the 0–50% grid.

$\gamma = 0$. This decomposition guides the counterfactuals in the main text: changing R&D incentives affects g_Y through the endogenous R&D component, while changing product-market allocations changes the output matrix $\boldsymbol{Q}$ and hence how a given knowledge-capital distribution maps into aggregate output.

B.8 Spillover-Intensity Sensitivity

Because β affects the counterfactuals through spillover growth, the recalibration of μ and δ, and the solved R&D policies, Table 12 propagates a rounded grid of beta values through the full structural calculation.

B.9 Product-Market Parameter Sensitivity

Table 13 varies the product-market parameter around the baseline value while keeping the central spillover parameter fixed at $\beta = 0.02$. For each value, the 2017 knowledge-capital state is recovered with the corresponding product-market matrix and the dynamic parameters are recalibrated to the same R&D-intensity and observed-growth targets. The main quantitative ordering is stable: the fixed-product-market planner gain remains around one half percent, the optimal uniform subsidy remains close to one third, and the full social-planner gain remains much larger.

B.10 Discount-Rate Sensitivity

Table 14 varies the discount rate and recalibrates the dynamic parameters to the same R&D-intensity and observed-growth targets. Lower discount rates put more weight on

Table 13: Product-Market Parameter Sensitivity

α	μ	δ	Median SMR/PMR	CS welfare (CC=100)	SS output (CC=100)	SS welfare (CC=100)	Subsidy rate (%)	Subsidy welfare (CC=100)
0.08	0.0527	0.0111	1.318	100.54	109.86	109.52	33	100.49
0.12	0.0526	0.0112	1.305	100.50	109.29	108.99	33	100.46
0.16	0.0521	0.0113	1.296	100.46	108.79	108.49	32	100.42

Notes: For each α, the 2017 knowledge-capital state is recovered from accounting profits using the corresponding product-market matrix, and μ and δ are recalibrated in the competitive equilibrium to match aggregate R&D intensity of 2.6% and observed-state g_Y of 1.5%. The spillover parameter is fixed at $\beta = 0.02$ and the discount rate at $\rho = 0.10$. Median SMR/PMR is the median of $\text{SMR}_i(z)/\text{PMR}_i(z)$ among firms with positive private and social marginal R&D returns. CS welfare, SS output, SS welfare, and subsidy welfare are normalized to no-subsidy competition at the same α. The subsidy rate is the welfare-maximizing uniform R&D subsidy on the 0–50% grid with 1 percentage point spacing. All scenario and subsidy-grid solutions are stable.

Table 14: Discount-Rate Sensitivity

ρ	μ	δ	Min. stability $\rho - 2g_{\max}$	CS welfare (CC=100)	SS welfare (CC=100)	Subsidy rate (%)	Subsidy welfare (CC=100)
5.0	0.0270	0.0098	0.0026	101.83	108.43	46	101.50
7.5	0.0397	0.0105	0.0336	100.70	108.44	36	100.64
10.0	0.0526	0.0112	0.0556	100.50	108.99	33	100.46

Notes: For each discount rate, μ and δ are recalibrated in the competitive 2017 equilibrium to match aggregate R&D intensity of 2.6% and observed-state g_Y of 1.5%. Welfare columns are normalized to the competitive equilibrium at the same ρ. Min. stability is the minimum of $\rho - 2g_{\max}$ across solved main scenarios, where $g_{\max}$ is the largest real eigenvalue of the closed-loop transition matrix Φ; positive values indicate finite discounted values. The subsidy rate is the welfare-maximizing uniform R&D subsidy on the 0–50% grid, and subsidy welfare is normalized to no-subsidy competition at the same ρ. All reported scenarios deliver finite Riccati/Lyapunov objects; the low-ρ case has the smallest stability margin.

future growth and therefore raise the welfare value of R&D interventions, but they also reduce the stability margin in the discounted Lyapunov equations. The baseline value $\rho = 10\%$ is conservative for the fixed-product-market R&D gains and keeps all reported scenarios well inside the finite-value region.

Appendix C Data, Implementation, and Numerical Checks

C.1 Sample Coverage and Network Overlap

Table 15 reports two compact summaries for the 2017 baseline sample. The first block compares the 757-firm quantitative sample with the relevant patenting-public-firm reference universe. The second block summarizes the raw overlap between the product-market and technology-spillover networks used in the estimated economy.

Table 15: Sample Coverage and Network Overlap in the 2017 Baseline

Statistic	Estimate	Firms
Baseline firms	757	757
Reference-universe firms	1,111	1,111
Share of reference-universe sales	90.7%	757/1,111
Share of reference-universe observed R&D	97.6%	757/1,111
Share of reference-universe gross profits	94.4%	757/1,111
Share of reference-universe DISCERN patents, 2015–2019	99.5%	757/1,111
Product centrality × technology-source centrality	0.22	757
Sales-weighted product centrality × technology-source centrality	0.27	757

Notes: The reference universe consists of 2017 Compustat firms with positive gross profits and at least one DISCERN-linked patent in the 2015–2019 grant window. Coverage rows report the baseline sample's share of the corresponding reference-universe aggregate. Network-overlap rows report Spearman correlations across the 757 baseline firms. Product centrality is off-diagonal product-market strength; technology-source centrality is off-diagonal column strength in the row-receiver spillover matrix. Sales-weighted centralities weight destination firms by observed sales.

C.2 Computational Scale

For the competitive dynamic problem, we solve the coupled system of Riccati equations by iterating on the coefficient matrices of firms' quadratic value functions. The implementation stores the firm-level value matrices as a three-dimensional array, recomputes the policy matrix $\widetilde{\boldsymbol{X}}$ and closed-loop transition matrix $\boldsymbol{\Phi}$ at each iteration, and updates firm pages using blockwise vectorized matrix products. For the monopoly and social-planner dynamic problems, the corresponding dynamic block is a single Riccati equation because one decision maker internalizes all R&D choices. After solving the R&D rule, we compute household welfare and aggregate producer value from continuous-time Lyapunov equations under the induced closed-loop law of motion.

When the number of firms is n, each firm has an $n \times n$ matrix of value-function coefficients, $\boldsymbol{X}^i$. The competitive dynamic problem therefore stores one matrix page per firm, or n^3 coefficients in total. In the 2017 quantitative baseline, $n = 757$, so this amounts to about 434 million coefficients. The main arithmetic cost comes from Riccati updates such as $\boldsymbol{\Phi}^T \boldsymbol{X}^i$ and $\boldsymbol{X}^i \boldsymbol{\Phi}$: each dense matrix product costs order $O(n^3)$ for one firm page, and updating all n pages gives an $O(n^4)$ operation.

Within existing endogenous growth frameworks, solving a dynamic oligopoly problem involving all publicly traded U.S. firms with patents is computationally challenging. With a discretized firm state for each firm, the state space grows exponentially in the number of firms, at order $O(k^n)$ for k grid points per firm. For instance, Cavenaile et al. (2026) introduce a dynamic oligopoly into a Schumpeterian growth model, similar to ours.

Table 16: Computational Complexity and Scale

Model	Computational complexity	Number of firms	Firm state space
Cavenaile et al. (2026)	$O(k^n)$	4	Six grid points per firm
Our model	$O(n^4)$	418–802	Continuous

Notes: Complexity entries describe the core state-space or dense-matrix operations that determine scalability. In our algorithm, storage of the competitive value-function coefficients scales as $O(n^3)$ and the $O(n^4)$ term refers to dense matrix operations within a Riccati update.

However, due to the curse of dimensionality, they analyze oligopolies with a maximum of four firms. In contrast, our model has a computational complexity of order $O\left(n^4\right)$ and preserves a continuous state space. This tractability is what makes it possible to evaluate counterfactual allocations for the full empirical network rather than for a small calibrated industry. Table 16 summarizes this computational contrast and the empirical scale of our application.

C.3 Riccati Solver, Selection, and Numerical Checks

For the competitive dynamic problem, the numerical algorithm used in the reported quantitative exercises iterates on the coefficient matrices of firms' quadratic value functions in the deterministic specification, $\gamma = 0$. Let k denote the iteration index. Given the current firm-level value matrices $\boldsymbol{X}_k^1, \ldots, \boldsymbol{X}_k^n$, we recover $\widetilde{\boldsymbol{X}}_k$, construct the implied closed-loop matrix $\boldsymbol{\Phi}_k$, and update the matrices using the forward-Euler step

$$\frac{\boldsymbol{X}_{k+1}^i - \boldsymbol{X}_k^i}{\Delta} = \boldsymbol{Q}_i - \mu^2 \boldsymbol{X}_{i,k}^i \left(\boldsymbol{X}_{i,k}^i\right)^T + \left(\boldsymbol{\Phi}_k - \frac{\rho}{2}\boldsymbol{I}\right)^T \boldsymbol{X}_k^i + \boldsymbol{X}_k^i \left(\boldsymbol{\Phi}_k - \frac{\rho}{2}\boldsymbol{I}\right).$$

Here Δ is a pseudo-time step for the fixed-point iteration, not a model time step. The vector $\boldsymbol{X}_{i,k}^i$ is the ith column of firm i's value-function matrix at iteration k, equivalently the transpose of the ith row of $\widetilde{\boldsymbol{X}}_k$. The stochastic Riccati equations in the model include the diffusion term shown in Equation (11); accordingly, the reported solver diagnostics evaluate the deterministic Riccati system used in the quantitative analysis. The implementation follows these steps:

1. Initialize $\boldsymbol{X}_0^i = \boldsymbol{0}$ for all firms.

2. After each update, recover $\widetilde{\boldsymbol{X}}_k$ from the firm-level value matrices and construct the closed-loop transition matrix $\boldsymbol{\Phi}_k$.

3. Update the competitive firm-value matrices in page blocks, which keeps the memory

footprint manageable while using vectorized dense matrix products within each block.

4. Stop when the maximum absolute change in $\widetilde{X}_k$ falls below the convergence tolerance. The pseudo-time step, tolerance, maximum iteration count, and page-block size are recorded in the replication code.

5. As a check separate from the convergence criterion, evaluate stability using the largest real eigenvalue of $\mathbf{\Phi}_k - \rho \boldsymbol{I}/2$.

6. For the monopoly and social-planner benchmarks, solve the corresponding single-agent Riccati equation with the same convergence criterion, using the path-flow drift term $\boldsymbol{X}\mathbf{\Phi} + \mathbf{\Phi}^T \boldsymbol{X}$.

The initialization also fixes the equilibrium-selection rule for the coupled Riccati system. For the no-subsidy competitive solves, the algorithm starts from zero value-function coefficients, $\boldsymbol{X}_0^i = \mathbf{0}$, and reports the stabilizing fixed point reached by the pseudo-time Riccati flow. This selected solution can be interpreted as the limit of finite-horizon LQ feedback games with zero terminal values when that limit exists (Cartigny and Michel, 2003). In the uniform-subsidy exercise, the implementation follows the same selected branch by using continuation over the subsidy grid: the solution at one grid point is used as the initial condition for the next grid point. Thus the reported subsidy counterfactuals hold fixed the branch-selection rule rather than searching across all possible stabilizing Riccati fixed points.

Table 17 reports numerical checks for the selected Riccati solutions. The residuals are scale-adjusted max norms of the algebraic Riccati equations evaluated at the reported solution, and the stability columns report the deterministic stability margin associated with Definition 2. These checks document residual accuracy and stability of the computed solution; they do not establish global uniqueness of the coupled Riccati fixed point.

Table 18 adds a local restart check for the 2017 competitive benchmark. The random starts perturb all firm-level value-function pages and return to the same selected solution up to numerical tolerance. This check does not detect another nearby solution in the tested restart battery, but it is not a proof that no other stabilizing solution exists.

C.4 Interior-Benchmark Non-Negativity Checks

The main scenario comparison uses the unconstrained interior static rules, $\boldsymbol{q}^u(\boldsymbol{z}) = \boldsymbol{N}\boldsymbol{z}$, and the unconstrained linear R&D policies, $\boldsymbol{x}^u(\boldsymbol{z}) = \mu \widetilde{\boldsymbol{X}} \boldsymbol{z}$, evaluated at the observed 2017

Table 17: Riccati Equation and Stability Diagnostics

Scenario	Riccati residual (relative max norm)	$\max \mathrm{Re}(\Phi - \rho I/2)$	$\rho - 2g_{\max}$	Saved $\widetilde{X}$ diff.
CC	1.7×10^{-6}	-0.0376	0.0753	0
CM	1.8×10^{-6}	-0.0370	0.0740	–
CS	1.6×10^{-6}	-0.0320	0.0640	–
MM	1.4×10^{-6}	-0.0369	0.0738	–
SS	1.9×10^{-6}	-0.0280	0.0561	–
CC_s_33	1.6×10^{-6}	-0.0343	0.0686	2.4×10^{-7}

Notes: The Riccati residual is the scale-adjusted max norm of the deterministic algebraic equation evaluated at the reported solution. $g_{\max}$ is the largest real eigenvalue of Φ. Negative values of $\max \mathrm{Re}(\Phi - \rho I/2)$, equivalently positive $\rho - 2g_{\max}$, indicate the deterministic stabilizing condition used in Assumption 1. The competitive rows, CC and the uniform-subsidy row, are re-solved only to recover the omitted firm-level Riccati pages; the last column reports the max absolute difference between the recovered $\widetilde{X}$ and the saved result. Single-agent rows compute the residual directly from the saved $\widetilde{X}$, so the last column is not applicable.

Table 18: Riccati Restart Checks for the 2017 Competitive Benchmark

Initial condition	Runs	Conv.	Worst error	Max $\widetilde{X}$ dist. (relative)	Max policy dist. (relative)
zero	1	1	5.1×10^{-7}	0	0
random scale 10^{-5}	2	2	5.1×10^{-7}	1.6×10^{-9}	5.1×10^{-9}
random scale 10^{-4}	2	2	5.1×10^{-7}	4.6×10^{-9}	1.1×10^{-8}

Notes: The table re-solves the 2017 competitive Riccati system from zero and from random symmetric firm-level value-function pages. Random scales are relative to $\max_{ij} |\widetilde{X}_{ij}|$ in the saved selected solution. Distances are relative Frobenius norms for $\widetilde{X}$ and relative Euclidean distances for the observed-state R&D policy. The restart exercise checks attraction back to the selected solution; it is not a proof of global uniqueness.

knowledge-capital vector. Because these rules can assign negative quantities or R&D effort in some counterfactuals, this subsection reports sign and projection checks for the reported interior benchmark; it does not recompute constrained static or dynamic equilibria.

The main scenario table uses two reporting conventions. Reported R&D expenditure is computed using the positive-effort resource-cost convention, $\sum_i \max\{0, x_i^u(z)\}^2$, while observed-state g_Y is computed from the signed linear drift implied by the solved policy at the observed state. The positive-effort convention prevents negative effort components from contributing positive squared costs to the expenditure aggregate.

Static Quantity Checks

Table 19 reports negative interior static quantities at the observed 2017 knowledge-capital vector. There are no negative quantities when the static allocation is competitive (CC, CM,

Table 19: Negative Interior Static Quantities at the Observed 2017 State

Scenario	Neg. firms	Neg. $\lvert q\rvert$ (%)	Neg. q^2 (%)	Sales (%)	Obs. R&D (%)
CC	0/757	0.00	0.00	0.00	0.00
CM	0/757	0.00	0.00	0.00	0.00
CS	0/757	0.00	0.00	0.00	0.00
MM	69/757	1.08	0.07	0.13	0.62
SS	234/757	7.91	0.79	1.03	2.58

Notes: The table evaluates the unconstrained interior static mapping $q^u(z) = Nz$ at the observed 2017 knowledge-capital vector. Negative $|q|$ is $\sum_{i:q_i^u<0} |q_i^u| / \sum_i |q_i^u|$, and negative q^2 is $\sum_{i:q_i^u<0} (q_i^u)^2 / \sum_i (q_i^u)^2$. Sales and observed R&D are observed 2017 shares accounted for by firms with negative interior static quantities. The table is a sign check and does not recompute an active-set allocation with $q_i \geq 0$.

Table 20: Static Nonnegative-Quantity QP Robustness at the Observed 2017 State

Scenario	Interior output	$q \geq 0$ QP output	Output change	Zero-quantity firms
MM	96.33	96.27	-0.06	77/757
SS	109.29	108.82	-0.47	283/757

Notes: Output is evaluated at the observed 2017 knowledge-capital vector and normalized by competitive-equilibrium output. The interior column uses the unconstrained static mapping in the main scenario table. The $q \geq 0$ QP column solves the corresponding static concave quadratic program with nonnegative quantities, holding the 2017 state fixed. The maximum KKT violation is 1.82×10^{-12} for MM and 9.09×10^{-13} for SS. This table is a static-block robustness check and does not recompute the dynamic Riccati controller.

and CS), so the constrained R&D-control counterfactuals do not raise this issue. Negative quantities arise only in the full static reallocation benchmarks: 69 firms in MM and 234 firms in SS. The affected firms are economically small in the quadratic quantity aggregate: they account for 0.07 percent of q^2 in MM and 0.79 percent in SS. These checks measure the extent of the interior-benchmark sign violation; they do not impose $q_i \geq 0$ or recompute active-set static matrices.

Table 20 solves the corresponding static nonnegative-quantity QPs for MM and SS at the same 2017 state. This check addresses the static sign restriction directly while leaving the dynamic policy and Riccati controller unchanged. The constrained-QP output values remain close to the interior benchmark values: output falls from 96.33 to 96.27 in MM and from 109.29 to 108.82 in SS.

R&D Effort Checks

Table 21 reports the number and severity of negative unconstrained R&D effort at the observed 2017 knowledge-capital vector. The competitive benchmark has no negative R&D

Table 21: Negative Unconstrained R&D Effort at the Observed 2017 State

Scenario	Neg. firms	Neg. R&D cost (%)	Sales (%)	Obs. R&D (%)
CC	0/757	0.00	0.00	0.00
CM	421/757	5.12	5.27	6.93
CS	23/757	0.01	0.01	0.07
MM	228/757	0.96	1.36	1.68
SS	111/757	0.17	0.27	0.72

Notes: Negative R&D cost share is $\sum_{i:x_i<0} x_i^2 / \sum_i x_i^2$. Sales and observed R&D are shares of observed 2017 totals accounted for by firms with negative unconstrained R&D effort.

Table 22: Projected-Positive Robustness of the 2017 Scenario Comparison

Scenario	Signed growth (%)	Projected growth (%)	Difference (pp)
CC	1.48	1.48	0.0000
CM	1.33	1.33	0.0014
CS	1.76	1.76	-0.0001
MM	1.37	1.37	-0.0007
SS	1.83	1.83	-0.0029

Notes: The projected-positive policy sets $x^+(z) = \max\{0, x^u(z)\}$ and uses x^+ consistently in both R&D expenditure and the knowledge drift. This adjustment holds the solved policy fixed and does not recompute a constrained equilibrium.

effort. Negative effort appears in some counterfactuals, especially CM, but the affected firms account for a small share of the R&D cost aggregate used in the scenario table. In CM, for example, 421 firms have negative unconstrained effort, while their negative R&D cost share is 5.12 percent, their sales share is 5.27 percent, and their observed-R&D share is 6.93 percent. The other counterfactuals have much smaller negative R&D cost shares.

Projected-Positive Growth Robustness

To gauge the importance of negative R&D components for observed-state growth, we set $\boldsymbol{x}^+(z) = \max\{\mathbf{0}, \boldsymbol{x}^u(z)\}$ componentwise and recompute both R&D expenditure and the knowledge drift using this fixed projected policy. This is a projection exercise, not a constrained equilibrium. Table 22 shows that observed-state g_Y changes by less than 0.003 percentage points in every scenario.

Transition-Path Sign and Welfare Robustness

The observed-state checks do not by themselves cover the infinite-horizon welfare object, which integrates payoffs along the induced transition path. We therefore simulate

Table 23: Transition-Path Sign Checks from the Observed 2017 State

Scenario	First $x < 0$ (years)	First $q < 0$ (years)	Max neg. R&D cost (%)	Max neg. q^2 (%)
CC	24.5	33.3	0.004	0.002
CM	0.0	30.9	6.624	0.004
CS	0.0	23.7	0.084	0.011
MM	0.0	0.0	0.958	0.125
SS	0.0	0.0	0.767	1.089
33% subsidy	15.2	26.6	0.012	0.007

Notes: The table simulates deterministic closed-loop transition paths for 50 years from the observed 2017 knowledge-capital vector. The 33% subsidy row uses the solved welfare-maximizing uniform-subsidy equilibrium. No scenario has negative knowledge capital along the simulated path. Negative R&D cost is $\sum_{i:x_i<0} x_i^2 / \sum_i x_i^2$; negative q^2 is $\sum_{i:q_i<0} q_i^2 / \sum_i q_i^2$. A value of 0.0 in a first-violation column means the corresponding sign violation is already present at the observed 2017 state.

deterministic closed-loop paths from the observed 2017 state and record the first time at which knowledge capital, R&D effort, or static quantities become negative. Table 23 reports the main sign checks. Knowledge capital remains positive in all scenarios over the 50-year horizon. The competitive benchmark is interior at the observed state and develops only late and economically tiny violations: the first negative R&D effort appears after 24.5 years and the first negative static quantity after 33.3 years. The constrained and full-control counterfactuals have some negative controls or quantities already at the observed state, but the corresponding negative mass is small relative to the aggregate objects used in the scenario table.

We apply the same projected-positive policy along deterministic transition paths and integrate household payoffs. Table 24 shows that welfare values are nearly unchanged: the largest movement is CM, at 0.1320 percentage points, and the scenario ranking is unchanged.

C.5 BGP Positivity Checks

This subsection concerns the long-run BGP interpretation of the solved closed-loop systems. It is distinct from the observed-state counterfactual comparisons in the main text, which evaluate welfare and g_Y at the 2017 knowledge-capital distribution.

Baseline BGP Positivity Check

The long-run sign restriction concerns the dominant eigenvector of the unconstrained closed-loop system, not the observed-state g_Y column in Table 8. The dominant eigenpair

Table 24: Projected-Positive R&D Welfare Robustness

Scenario	Signed-policy welfare (CC=100)	Projected-positive welfare (signed CC=100)	Difference (pp)
CC	100.0000	100.0001	0.0001
CM	98.9113	99.0433	0.1320
CS	100.4965	100.4959	-0.0005
MM	95.6637	95.6833	0.0196
SS	108.9720	108.9605	-0.0115
33% subsidy	100.4514	100.4514	0.0001

Notes: The projected-positive policy sets negative unconstrained R&D effort to zero and uses the resulting policy consistently in the knowledge drift and in R&D resource costs. The calculation integrates household payoffs for 100 years and adds a fixed-active-set terminal tail; the terminal-tail share is below 0.052 percent in every row. The projected-positive policy is a feasible-policy robustness check around the reported unconstrained policy, not a recomputed constrained Markov equilibrium.

solves

$$\boldsymbol{\Phi}\bar{\boldsymbol{z}} = g\bar{\boldsymbol{z}}, \qquad \boldsymbol{\Phi} = \boldsymbol{\Omega} - \delta \boldsymbol{I} + \mu^2 \widetilde{\boldsymbol{X}}. \tag{50}$$

Although $\boldsymbol{\Omega}$ is nonnegative under the row-receiver spillover convention, $\widetilde{\boldsymbol{X}}$ contains strategic R&D policy feedbacks and can have negative off-diagonal elements. Hence $\boldsymbol{\Phi}$ is not generally a Metzler matrix and need not preserve the positive orthant; a positive dominant eigenvector is therefore not guaranteed by Perron–Frobenius arguments. After orientation toward the observed knowledge-capital vector and normalization by $\sum_i |\bar{z}_i| = 1$, $\bar{\boldsymbol{z}}$ is the candidate cross-sectional knowledge-capital distribution on the BGP. If it has both positive and negative entries, the candidate path would assign negative knowledge capital to some firms, so it cannot be interpreted as a positive BGP. We therefore do not use the associated firm-level BGP R&D policies or depreciation-bound calculations as economic results. This restriction is specific to the asymptotic BGP calculation; Table 8 is evaluated at the observed 2017 state. Table 25 reports this positivity check for the five counterfactual scenarios.

Technology Spillover-Floor Check

The baseline BGP feasibility results show that the unconstrained dominant eigenvectors are mixed-sign in all five 2017 scenarios. To gauge whether this long-run sign issue is driven by the sparse allocation of empirical spillover links, we compute one perturbation for the competitive benchmark. The perturbation preserves each recipient firm's total spillover

Table 25: Positive-State Feasibility of the Unconstrained Dominant BGP

Scenario	Neg. $\bar{z}$	Neg. $\lvert\bar{z}\rvert$ share (%)
CC	524/757	47.9
CM	430/757	49.0
CS	384/757	49.3
MM	458/757	49.0
SS	286/757	50.6

Notes: The dominant eigenvector is oriented toward the observed 2017 knowledge-capital vector and normalized by $\sum_i |\bar{z}_i| = 1$. Since no scenario has a positive dominant state, firm-level BGP R&D non-negativity and depreciation lower-bound calculations are not economically interpreted.

exposure but spreads a fraction λ uniformly across other source firms:

$$\Omega_{ij}^{(\lambda)} = (1-\lambda)\Omega_{ij} + \lambda \frac{\sum_{k \neq i} \Omega_{ik}}{n-1}, \qquad j \neq i, \tag{51}$$

with the diagonal unchanged.

For $\lambda = 0.15$, the competitive benchmark has a positive oriented dominant eigenvector. Table 26 compares deterministic transitions from the same observed 2017 state under the baseline competitive system and this spillover-floor perturbation. The perturbation changes the instantaneous drift by about 2.7 percent in relative ℓ_2 norm, while firm-level growth rates, aggregate R&D, and continuation values remain close over the reported horizons. This exercise is not a re-estimated model and is not used in the main counterfactuals; it only checks that the observed-state comparison is not mechanically driven by the mixed-sign asymptotic BGP.

Table 26: Observed-State Transition Comparison for the CC Spillover-Floor Benchmark

Years	Normalized z distance (ℓ_1)	Firm-growth corr.	Growth diff. (pp)	R&D ratio	Welfare ratio
0	0.0000	0.9983	-0.045	0.996	0.995
1	0.0002	0.9983	-0.045	0.996	0.994
5	0.0008	0.9982	-0.046	0.996	0.993
10	0.0016	0.9982	-0.045	0.995	0.990
20	0.0030	0.9980	-0.042	0.994	0.987
50	0.0062	0.9962	-0.028	0.992	0.978

Notes: The table compares deterministic transitions from the same observed 2017 state under baseline CC and the row-sum-preserving $\lambda = 0.15$ spillover-floor benchmark. The state distance is the ℓ_1 distance between states normalized by $\sum_i |z_i(t)|$. Firm-growth corr. is the cross-firm correlation of instantaneous knowledge-growth rates, $(\boldsymbol{\Phi} \boldsymbol{z}(t))_i / z_i(t)$, between the two paths. Growth diff. is the benchmark path minus the baseline path in percentage points. R&D is the ratio of R&D expenditure to the baseline path at the same horizon. Welfare is the ratio of continuation values, $V(\boldsymbol{z}(t)) = \boldsymbol{z}(t)^T \boldsymbol{X} \boldsymbol{z}(t)$, evaluated at the same horizon.